

NOTE: These are just Practice Problems. This is NOT meant to look just like the test, and it is NOT the only thing that you should study. Make sure you know all the material from the notes, quizzes, suggested homework and the corresponding chapters in the book.

Unit 1

1 Pick some learning tasks. Describe it informally in a paragraph in English. Now describe it by stating as precisely as possible the task, performance measure, and training experience. Finally, propose a target function to be learned and a target representation. Discuss the main tradeoffs you considered in formulating this learning task.

2 You are working on a Face Recognition Problem, and you got the following result.

	Nearest Neighbor	Decision Tree	Neural Network	Support Vector Machine
Accuracy on training data	70%	80%	85%	75%
Accuracy on testing data	65%	70%	60%	75%

Which algorithm is the best algorithm?

Which algorithm is overfitting the most?

3 For the given three classes the predictions are tabulated in a confusion matrix. Find the f1 score for each class. Actual/Predicted

	A	B	C
A	0	0	2
B	1	3	1
C	0	0	3

4 With a neat diagram, discuss the process of designing a learner.

	<pre> graph TD A[Experiences] --> B[Training Experiences] B --> C[Knowledge] C --> D[Knowledge representation] D --> E[M/c learning algorithm] E --> F[Learner/Model] </pre> Choose the training experience: select experiences that have variance (capture most of the possible scenario) Choose the target function: determine exactly what type of knowledge will be learned Choose how to represent the target function: choose a representation to describe the knowledge (table base, rule based, hierarchical, polynomial, ANN) Choose a learning algorithm to infer the target function: choose the proper learning algorithm to build knowledge from experience										
5	Distinguish between Supervised and Reinforced Learning.										
	<table border="1"> <thead> <tr> <th>Supervised</th><th>Reinforced</th></tr> </thead> <tbody> <tr> <td>Works on existing or given samples data or samples</td><td>Works on interacting with the environment</td></tr> <tr> <td>Assets are depreciable</td><td>Liabilities are non-depreciable</td></tr> <tr> <td>Deperated with interactive software systems or applications</td><td>Supports and works better in artificial intelligence where human interaction is prevalent</td></tr> <tr> <td>Many open source projects are evolving of development in this area</td><td>More useful in Artificial Intelligence</td></tr> </tbody> </table>	Supervised	Reinforced	Works on existing or given samples data or samples	Works on interacting with the environment	Assets are depreciable	Liabilities are non-depreciable	Deperated with interactive software systems or applications	Supports and works better in artificial intelligence where human interaction is prevalent	Many open source projects are evolving of development in this area	More useful in Artificial Intelligence
Supervised	Reinforced										
Works on existing or given samples data or samples	Works on interacting with the environment										
Assets are depreciable	Liabilities are non-depreciable										
Deperated with interactive software systems or applications	Supports and works better in artificial intelligence where human interaction is prevalent										
Many open source projects are evolving of development in this area	More useful in Artificial Intelligence										

	Many algorithms exist in using this learning	Neither supervised nor unsupervised algorithms are used
	Runs on any platform or with any applications	Runs with any hardware or software devices
6	You are assigned a new project which involves helping a food delivery company save more money. The problem is company's delivery teams aren't able to deliver food on time. As a result, their customers get unhappy. And, to keep them happy, they end up delivering food for free. Which machine learning algorithm can save them?	
	This is not a machine learning problem. This is a route optimization problem. A machine learning problem consists of three things: 1. There exists a pattern. 2. You cannot solve it mathematically (even by writing exponential equations). 3. You have data on it. Always look for these three factors to decide if machine learning is a tool to solve a particular problem.	
7	What is machine learning? List four types of features that can be used in machine learning. Illustrate the four interrelated components.	
	A computer program is said to Learn from experience E with respect to Some class of tasks T and performance measure P, If its performance of tasks in T, as measured by P, improves with experience E. Types of features: Continuous (real values),Categorical (A, B, AB, O),Ordinal (large, medium, small) Discrete (integer values It can be divided into four interrelated components: Data storage: utilizes observation, memory, and recall to provide a factual basis for further reasoning Abstraction: involves the translation of stored data into broader representations and concepts Generalization: uses abstracted data to create knowledge and inferences that drive action in new contexts Evaluation: provides a feedback mechanism to measure the utility of learned knowledge and inform potential improvements	
8	Compare ill-posed and well problems with a suitable example.	

	If the problem is well-posed, then it stands a good chance of solution on a computer using a stable algorithm. An ill-conditioned problem is indicated by a large condition number. An ill posed problem is one which doesn't need the three Hadamard Criteria for being well posed. These criteria are: - Having a solution - Having a unique solution - Having a solution that depends continuously on the parameters or input data. Examples of well-posed problems include the Dirichlet problem for Laplace's equation, and the heat equation with specified initial conditions. Generally, Inverse problems are often ill-posed. Examples of ill posed One simple equation of an ill posed problem is given by the equation $y' = (3/2) y^{1/3}$ with $y(0)=0$. Since the equation is $y(t) = \pm t^{3/2}$, the solution is not unique (it could be plus $t^{3/2}$ or it could be minus $t^{3/2}$). As this violates rule 2 of the Hadamard criteria, the problem is ill posed. Many inverse problems are ill-posed because either they don't have a solution everywhere their solution is not unique, or their solution is not stable (continuous). Explain well-posed learning problems. Give any three related examples which you think are well posed learning problems.
	A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E. To have a well-defined learning problem, three features needs to be identified: 1. The class of tasks 2. The measure of performance to be improved 3. The source of experience Examples 1. Checkers game: A computer program that learns to play checkers might improve its performance as measured by its ability to win at the class of tasks involving playing checkers games, through experience obtained by playing games against itself. A checkers learning problem: ● Task T: playing checkers ● Performance measure P: percent of games won against opponents ● Training experience E: playing practice games against itself 2. A handwriting recognition learning problem: ● Task T: recognizing and classifying handwritten words within images ● Performance measure P: percent of words correctly classified

	● Training experience E: a database of handwritten 3. A robot driving learning problem: ● Task T: driving on public four-lane highways using vision sensors ● Performance measure P: average distance travelled before an error (as judged by human overseer) ● Training experience E: a sequence of images and steering commands recorded while observing a human driver
9	Design a learning system for any learning task and describe i)How to choose training experience. ii)How to choose the target function. iii)How to choose representation for the target function. iv) Explain function approximation algorithm?
	The basic design issues and approaches to machine learning are illustrated by designing a program to learn to play checkers, with the goal of entering it in the world checkers tournament 1. Choosing the Training Experience 2. Choosing the Target Function 3. Choosing a Representation for the Target Function 4. Choosing a Function Approximation Algorithm 1. Estimating training values 2. Adjusting the weights 5. The Final Design 1. Choosing the Training Experience ● The first design choice is to choose the type of training experience from which the system will learn. ● The type of training experience available can have a significant impact on success or failure of the learner. <u>There are three attributes which impact on success or failure of the learner</u> 1. Whether the training experience provides direct or indirect feedback regarding the choices made by the performance system. <u>For example, in checkers game:</u> In learning to play checkers, the system might learn from direct training examples consisting of individual checkers board states and the correct move for each.

Indirect training examples consisting of the **move sequences** and **final outcomes** of various games played. The information about the correctness of specific moves early in the game must be inferred indirectly from the fact that the game was eventually won or lost.

Here the learner faces an additional problem of **credit assignment**, or determining the degree to which each move in the sequence deserves credit or blame for the final outcome. Credit assignment can be a particularly difficult problem because the game can be lost even when early moves are optimal, if these are followed later by poor moves. Hence, learning from direct training feedback is typically easier than learning from indirect feedback.

2. The degree to which the **learner controls the sequence of training examples**

For example, in checkers game:

The learner might depend on the **teacher** to select informative board states and to provide the correct move for each.

Alternatively, the learner might itself propose board states that it finds particularly confusing and ask the teacher for the correct move.

The learner may have complete control over both the board states and (indirect) training classifications, as it does when it learns by playing against itself with **no teacher present**.

3. How well it represents the **distribution of examples** over which the final system performance P must be measured

For example, in checkers game:

In checkers learning scenario, the performance metric P is the percent of games the system wins in the world tournament.

If its training experience E consists only of games played against itself, there is a danger that this training experience might not be fully representative of the distribution of situations over which it will later be tested.

It is necessary to learn from a distribution of examples that is different from those on which the final system will be evaluated.

2. Choosing the Target Function

The next design choice is to determine exactly what type of knowledge will be learned and how this will be used by the performance program.

Let's consider a checkers-playing program that can generate the legal moves from any board state.

The program needs only to learn how to choose the best move from among these legal moves. We must learn to choose among the legal moves, the most obvious choice for the type of information to be learned is a program, or function, that chooses the best move for any given board state.

1. Let **ChooseMove** be the target function and the notation is

$$\text{ChooseMove} : B \rightarrow M$$

which indicate that this function accepts as input any board from the set of legal board states B and produces as output some move from the set of legal moves M .

ChooseMove is a choice for the target function in checkers example, but this function will turn out to be very difficult to learn given the kind of indirect training experience available to our system

2. An alternative target function is an **evaluation function** that assigns a **numerical score**

to any given board state

Let the target function V and the notation

$$V : B \rightarrow R$$

which denote that V maps any legal board state from the set B to some real value. Intend for this target function V to assign higher scores to better board states. If the system can successfully learn such a target function V , then it can easily use it to select the best move from any current board position.

Let us define the target value $V(b)$ for an arbitrary board state b in B , as follows:

- If b is a final board state that is won, then $V(b) = 100$
- If b is a final board state that is lost, then $V(b) = -100$
- If b is a final board state that is drawn, then $V(b) = 0$
- If b is not a final state in the game, then $V(b) = V(b')$,

Where b' is the best final board state that can be achieved starting from b and playing optimally until the end of the game

3. Choosing a Representation for the Target Function

Let's choose a simple representation - for any given board state, the function c will be calculated as a linear combination of the following board features:

- x1: the number of black pieces on the board
- x2: the number of red pieces on the board
- x3: the number of black kings on the board
- x4: the number of red kings on the board
- x5: the number of black pieces threatened by red (i.e., which can be captured on red's next turn)
- x6: the number of red pieces threatened by black

Thus, learning program will represent as a linear function of the form

$$\hat{V}(b) = w_0 + w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + w_5x_5 + w_6x_6$$

Where,

- w0 through w6 are numerical coefficients, or weights, to be chosen by the learning algorithm.
- Learned values for the weights w1 through w6 will determine the relative importance of the various board features in determining the value of the board
- The weight w0 will provide an additive constant to the board value

4. Choosing a Function Approximation Algorithm

In order to learn the target function **f** we require a set of training examples, each describing a specific board state **b** and the training value **Vtrain(b)** for **b**.

Each training example is an ordered pair of the form (b, Vtrain(b)).

For instance, the following training example describes a board state **b** in which black has won the game (note x2 = 0 indicates that red has no remaining pieces) and for which the target function value Vtrain(b) is therefore +100.

$$((x_1=3, x_2=0, x_3=1, x_4=0, x_5=0, x_6=0), +100)$$

Function Approximation Procedure

1. Derive training examples from the indirect training experience available to the learner
2. Adjusts the weights w_i to best fit these training examples

1. Estimating training values

A simple approach for estimating training values for intermediate board states is to assign the training value of $V_{train}(b)$ for any intermediate board state b to be $\hat{V}(\text{Successor}(b))$

Where ,

- $\hat{V}$ is the learner's current approximation to V
- $\text{Successor}(b)$ denotes the next board state following b for which it is again the program's turn to move

Rule for estimating training values

$$V_{train}(b) \leftarrow \hat{V}(\text{Successor}(b))$$

2. Adjusting the weights

Specify the learning algorithm for choosing the weights w_i to best fit the set of training examples $\{(b, V_{train}(b))\}$

A first step is to define what we mean by the bestfit to the training data. One common approach is to define the best hypothesis, or set of weights, as that which minimizes the squared error E between the training values and the values predicted by the hypothesis.

$$E \equiv \sum_{\langle b, V_{train}(b) \rangle \in \text{training examples}} (V_{train}(b) - \hat{V}(b))^2$$

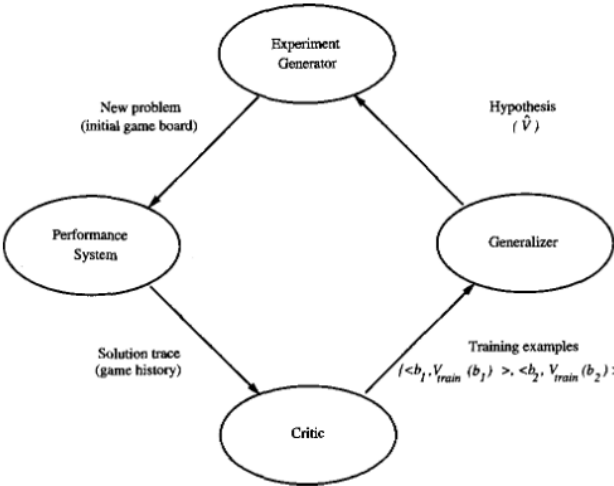
Several algorithms are known for finding weights of a linear function that minimize E . One such algorithm is called the **least mean squares, or LMS training rule**. For each observed training example it adjusts the weights a small amount in the direction that reduces the error on this training example

LMS weight update rule :- For each training example $(b, V_{train}(b))$ Use the current weights to calculate $\hat{V}(b)$
For each weight w_i , update it as

$$w_i \leftarrow w_i + \eta (V_{train}(b) - \hat{V}(b)) x_i$$

Here η is a small constant (e.g., 0.1) that moderates the size of the weight update. Working of weight update rule

- When the error $(V_{train}(b) - \hat{V}(b))$ is zero, no weights are changed.
- When $(V_{train}(b) - \hat{V}(b))$ is positive (i.e., when $\hat{V}(b)$ is too

	low), then each weight is increased in proportion to the value of its corresponding feature. This will raise the value of $\hat{V}(b)$, reducing the error. ● If the value of some feature x_i is zero, then its weight is not altered regardless of the error, so that the only weights updated are those whose features actually occur on the training example board. 5. The Final Design The final design of checkers learning system can be described by four distinct program modules that represent the central components in many learning systems  <pre> graph TD EG([Experiment Generator]) -- "New problem (initial game board)" --> PS([Performance System]) PS -- "Solution trace (game history)" --> C([Critic]) C -- "Training examples (b1, V_train(b1), ..., bn, V_train(bn))" --> G([Generalizer]) G -- "Hypothesis (V)" --> EG </pre>
10	Explain Concept learning as a search with an appropriate example.
	● Concept learning can be viewed as the task of searching through a large space of hypotheses implicitly defined by the hypothesis representation. ● The goal of this search is to find the hypothesis that best fits the training examples. Example: Consider the instances X and hypotheses H in the <i>EnjoySport</i> learning task. The attribute <i>Sky</i> has three possible values, and <i>AirTemp</i>, <i>Humidity</i>, <i>Wind</i>, <i>Water</i>, <i>Forecast</i> each have two possible values, the instance space X contains exactly $3 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 96$ distinct instances $5 \cdot 4 \cdot 4 \cdot 4 \cdot 4 \cdot 4 = 5120$ syntactically distinct hypotheses within H. Every hypothesis containing one or more "Φ" symbols represents the empty set of instances; that is, it classifies every instance as negative.

	1 + (4.3.3.3.3.3) = 973. Semantically distinct hypotheses																																								
11	Illustrate FIND-S Algorithm with an example.																																								
	<u>FIND-S Algorithm</u> 1. Initialize h to the most specific hypothesis in H 2. For each positive training instance x For each attribute constraint a_i in h If the constraint a_i is satisfied by x Then do nothing Else replace a_i in h by the next more general constraint that is satisfied by x 3. Output hypothesis h To illustrate this algorithm, assume the learner is given the sequence of training examples from the EnjoySport task <table><tr><th>Example</th><th>Sky</th><th>AirTemp</th><th>Humidity</th><th>Wind</th><th>Water</th><th>Forecast</th><th>EnjoySport</th></tr><tr><td>1</td><td>Sunny</td><td>Warm</td><td>Normal</td><td>Strong</td><td>Warm</td><td>Same</td><td>Yes</td></tr><tr><td>2</td><td>Sunny</td><td>Warm</td><td>High</td><td>Strong</td><td>Warm</td><td>Same</td><td>Yes</td></tr><tr><td>3</td><td>Rainy</td><td>Cold</td><td>High</td><td>Strong</td><td>Warm</td><td>Change</td><td>No</td></tr><tr><td>4</td><td>Sunny</td><td>Warm</td><td>High</td><td>Strong</td><td>Cool</td><td>Change</td><td>Yes</td></tr></table> • The first step of FIND-S is to initialize h to the most specific hypothesis in H $h = (\emptyset, \emptyset, \emptyset, \emptyset, \emptyset, \emptyset)$ • Consider the first training example $x_1 = \langle \text{Sunny Warm Normal Strong Warm Same} \rangle, +$ Observing the first training example, it is clear that hypothesis h is too specific. None of the "$\emptyset$" constraints in h are satisfied by this example, so each is replaced by the next <i>more general constraint</i> that fits the example $h_1 = \langle \text{Sunny Warm Normal Strong Warm Same} \rangle$ • Consider the second training example $x_2 = \langle \text{Sunny, Warm, High, Strong, Warm, Same} \rangle, +$ The second training example forces the algorithm to further generalize h, this time substituting a "?" in place of any attribute value in h that is not satisfied by the new example $h_2 = \langle \text{Sunny Warm ? Strong Warm Same} \rangle$	Example	Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport	1	Sunny	Warm	Normal	Strong	Warm	Same	Yes	2	Sunny	Warm	High	Strong	Warm	Same	Yes	3	Rainy	Cold	High	Strong	Warm	Change	No	4	Sunny	Warm	High	Strong	Cool	Change	Yes
Example	Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport																																		
1	Sunny	Warm	Normal	Strong	Warm	Same	Yes																																		
2	Sunny	Warm	High	Strong	Warm	Same	Yes																																		
3	Rainy	Cold	High	Strong	Warm	Change	No																																		
4	Sunny	Warm	High	Strong	Cool	Change	Yes																																		

- Consider the third training example
 $x_3 = \langle \text{Rainy, Cold, High, Strong, Warm, Change} \rangle, -$

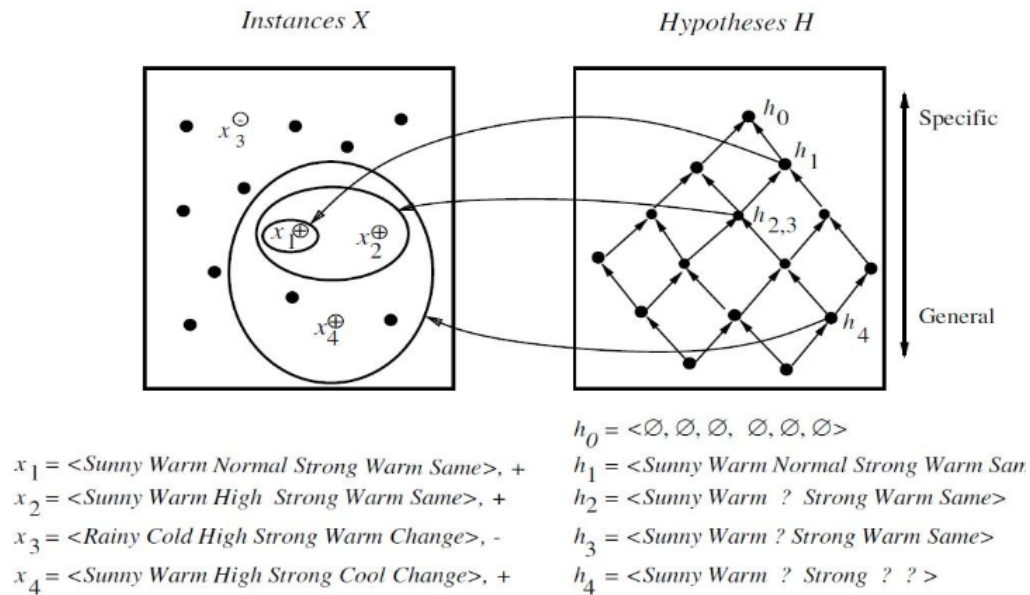
Upon encountering the third training the algorithm makes no change to h . The FIND-S algorithm simply ignores every negative example.

$h_3 = \langle \text{Sunny Warm ? Strong Warm Same} \rangle$

- Consider the fourth training example
 $x_4 = \langle \text{Sunny Warm High Strong Cool Change} \rangle, +$

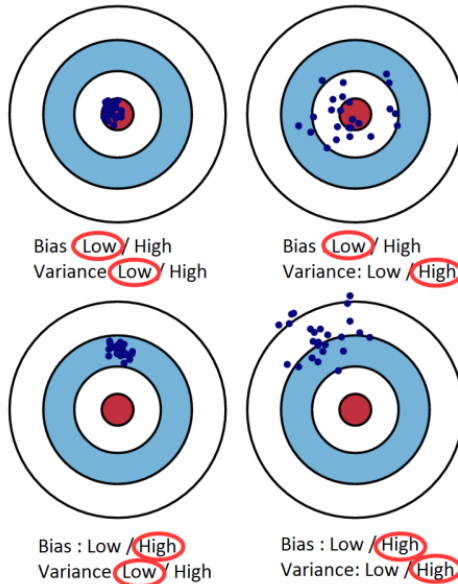
The fourth example leads to a further generalization of h

$h_4 = \langle \text{Sunny Warm ? Strong ? ?} \rangle$



12	Imagine you are working on a project which is a binary classification problem. You trained a model on the training dataset and got the below confusion matrix on validation dataset. <table><tr><td></td><th>Predicted: NO</th><th>Predicted: YES</th></tr><tr><th>Actual: NO</th><td>50</td><td>10</td></tr><tr><th>Actual: YES</th><td>5</td><td>100</td></tr></table> Based on the above confusion matrix, compute accuracy, true positive rate and false positive rate.		Predicted: NO	Predicted: YES	Actual: NO	50	10	Actual: YES	5	100
	Predicted: NO	Predicted: YES								
Actual: NO	50	10								
Actual: YES	5	100								
	Accuracy is ~0.91, True positive rate is ~0.95. The Accuracy (correct classification) is (50+100)/165 which is nearly equal to 0.91. The true Positive Rate is how many times you are predicting positive class correctly so true positive rate would be 100/105 = 0.95									
13	How do you ensure you're not overfitting with a model?									
	This is a simple restatement of a fundamental problem in machine learning: the possibility of overfitting training data and carrying the noise of that data through to the test set, thereby providing inaccurate generalizations. There are three main methods to avoid overfitting: 1- Keep the model simpler: reduce variance by considering fewer variables and parameters, thereby removing some of the noise in the training data. 2- Use cross-validation techniques such as k-folds cross-validation. 3- Use regularization techniques such as LASSO that penalize certain model parameters if they're likely to cause overfitting.									
14	To understand bias and variance, we will create a graphical visualization using a bullseye. Imagine that the center of the target is our true model (a model that perfectly predicts the correct values). As we move away from the bullseye, our predictions get worse and worse. Imagine we can repeat our entire model building process to get several separate hits on the target. Each hit represents an individual realization of our model, given the chance variability in the training data we gather. Sometimes we will get a good distribution of training data so we predict very well, and we are close to the bullseye, while sometimes our training data might be full of outliers or non-standard values resulting in poorer predictions. Consider these four different realizations resulting from a scatter of hits on the target. Characterize the bias and variance of the estimates of the following models on the data with respect to the true model as low or high by									

circling the appropriate entries below each diagram.

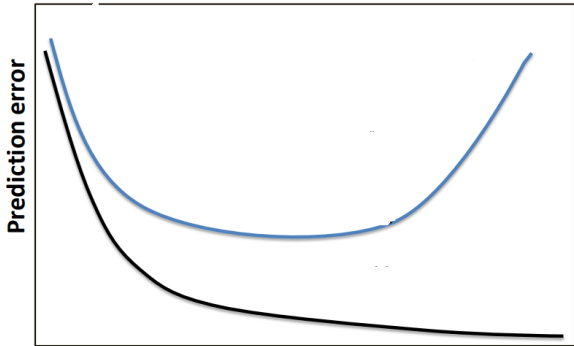
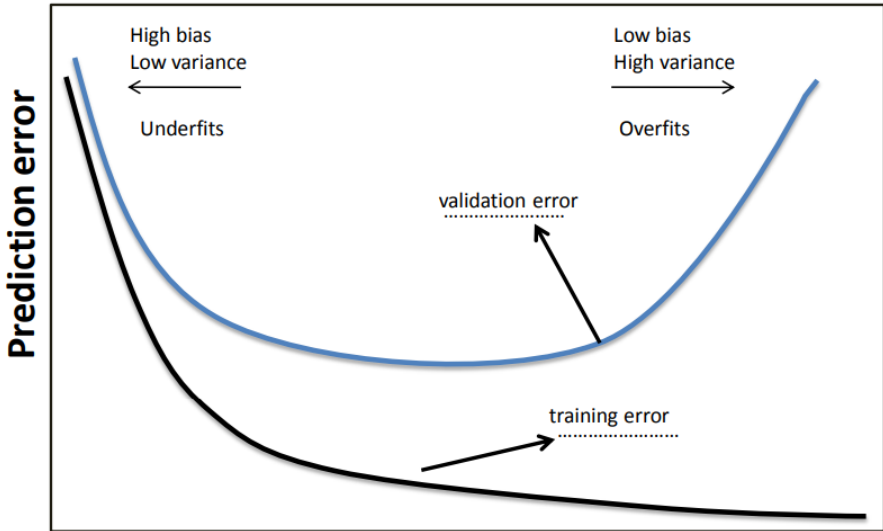


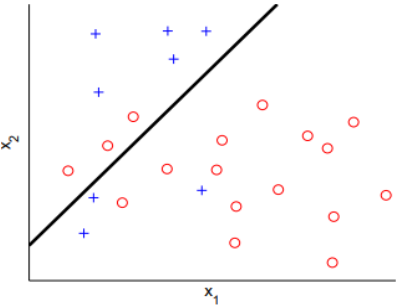
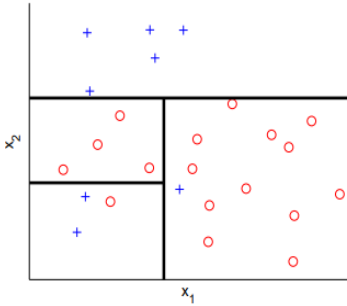
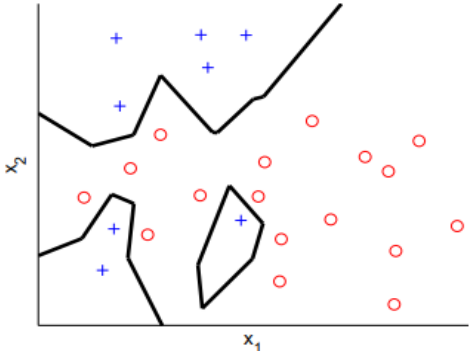
Explain what effect the following operations will have on the bias and variance of your model. Fill in one of ‘increases’, ‘decreases’ or ‘no change’ in each of the cells:

	Bias	Variance
Regularizing the weights in a linear/logistic regression model	increases	decreases
Increasing k in k-nearest neighbor models	increases	decreases
Pruning a decision tree (to a certain depth for example)	increases	decreases
Increasing the number of hidden units in an artificial neural network	decreases	increases
Using dropout to train a deep neural network	increases	decreases
Removing all the non-support vectors in SVM	no change	no change

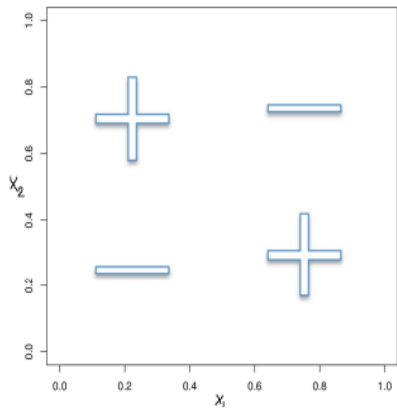
15

The following figure depicts training and validation curves of a learner with increasing model complexity.

	 Prediction error Model complexity Which of the curves is more likely to be the training error and which is more likely to be the validation error? Indicate on the graph by filling the dotted lines. In which regions of the graph are bias and variance low and high? Indicate clearly on the graph with four labels: “low variance”, “high variance”, “low bias”, “high bias”. In which regions does the model overfit or underfit? Indicate clearly on the graph by labeling “overfit” and “underfit”.
	 Prediction error Model complexity High bias Low variance ← Underfits Low bias High variance → Overfits validation error training error
16	A set of data points is generated by the following process: $Y = w_0 + w_1X + w_2X^2 + w_3X^3 + w_4X^4 + \epsilon$, where X is a real-valued random variable and ϵ is a Gaussian noise variable. You use two models to fit the data: Model 1: $Y = aX + b + \epsilon$ Model 2: $Y = w_0 + w_1X^1 + \dots + w_9X^9 + \epsilon$ 1. Model 1, when compared to Model 2 using a fixed number of training examples, has a bias which is: (a) Lower (b) Higher (c) The Same

	2. Model 1, when compared to Model 2 using a fixed number of training examples, has a variance which is: (a) Lower (b) Higher (c) The Same 3. Given 10 training examples, which model is more likely to overfit the data? (a) Model 1 (b) Model 2
	1. Higher, 2. Lower, 3. Model 2
	The following figures depict decision boundaries of classifiers obtained from three learning algorithms: decision trees, logistic regression, and nearest neighbor classification (in some order). Beside each of the three plots, write: name of the learning algorithm and number of mistakes it makes on the training data. + positive training examples ○ negative training examples   

	[Points: 4 pts] Name: ★ Logistic regression Number of mistakes: ★ 6 [Points: 4 pts] Name: ★ Decision tree Number of mistakes: ★ 2 [Points: 4 pts] Name: ★ k-nearest neighbor Number of mistakes: ★ 0
17	A classifier that attains 100% accuracy on the training set and 70% accuracy on the test set is better than a classifier that attains 70% accuracy on the training set and 75% accuracy on the test set.
	FALSE. The second classifier has better test accuracy which reflects the true accuracy, whereas the first classifier is overfitting.
18	Which of the following classifiers can perfectly classify the following data:



(a) Decision Tree (b) Logistic Regression (c) Gaussian Naive Bayes

Decision Tree only. Decision trees of depth 2 which first splits on X_1 and then on X_2 will perfectly classify it. Logistic regression leads to linear decision boundaries, hence cannot classify this data perfectly. Due to conditional independence requirement, it is not possible to fit a Gaussian that peaks at the labels of only one class and has no covariance between features, so Gaussian Naive Bayes cannot classify this data perfectly.

19

Let's say you are building a model that detects whether a person has diabetes or not. After the train-test split, you get a test set of length 100, out of which 70 data points are labelled positive (1), and 30 data points are labelled negative (0). Out of 70 actual positive data points, your model predicted 64 points as positive and 6 as negative. Out of 30 actual negative points, it predicted 3 as positive and 27 as negative. Draw the matrix for your test prediction and evaluate the model as smart model or dumb model.

	Actually Positive	Actually Negative
Predicted Positive	64	3
Predicted Negative	6	27

Note: In the notations, True Positive, True Negative, False Positive, & False Negative, notice that the second term (Positive or Negative) is denoting your prediction, and the first term denotes whether you predicted right or wrong. Based on the above matrix of diabetes detection model, we can calculate these ratios:

$$\text{TPR} = 91.4\%$$

	Precision measures the percentage of emails flagged as spam that were correctly classified—that is, the percentage of dots to the right of the threshold line that are green in Figure 1: $\text{Precision} = \frac{TP}{TP + FP} = \frac{8}{8 + 2} = 0.8$ Recall measures the percentage of actual spam emails that were correctly classified—that is, the percentage of green dots that are to the right of the threshold line in Figure 1: $\text{Recall} = \frac{TP}{TP + FN} = \frac{8}{8 + 3} = 0.73$
21	Describe Cures for Overfitting.
	If the model is overfitting, the most sensible approach is to reduce model complexity. This way, we can enable it to generalize better. In an attempt to reduce model flexibility, we can reduce the number of input features. To enable fewer feature combinations within the model, we can either manually select which features to keep or take advantage of a feature selection algorithm. Alternatively, we can apply regularization to suppress higher-order terms in our model. So, we keep all the features but limit the magnitude of their feature importance in the outcome. As expected, regularization works well with a lot of features that contain slightly helpful information. For instance, adding a dropout layer in neural networks is a form of regularization. Moreover, we can set a regularization term to limit the weights of a neural network. Early stopping is another type of regularization we can utilize for iterative models. The main idea is to stop model training when specific criteria are satisfied, such as when validation accuracy starts to decrease. Another possible approach to model fitting is to use more training examples to train the model to generalize better.
22	Describe Cures for Underfitting.
	To prevent underfitting, we need to ensure the model complexity. The first method that comes to mind is to obtain more training data. However, this is not an easy task for most problems. In such cases, we can bring data augmentation into service. So, we can increase the amount of data available by creating slightly modified synthetic copies of the data points at hand. Similarly, increasing the number of passes on the training data is a viable

	approach for iterative algorithms. Increasing the number of epochs in a neural network is a well-known practice to ensure model fitting. Another way to increase model complexity is to increase the size and number of model parameters. We can introduce engineered features from the dataset. For example, a product of numerical features or n parameter of n-grams generates new features. Alternatively, we can reduce regularization. Some implementations implicitly include default regularization parameters to overfitting. Checking the default parameters is a good start point. As we're trying to get out of a limited feature set, there's no need to introduce limiting terms into the model. Replacing the approach is another solution. For example, the selection of the kernel function in SVM determines the model complexity. Thus, the choice of kernel function might lead to overfitting or underfitting.
--	---

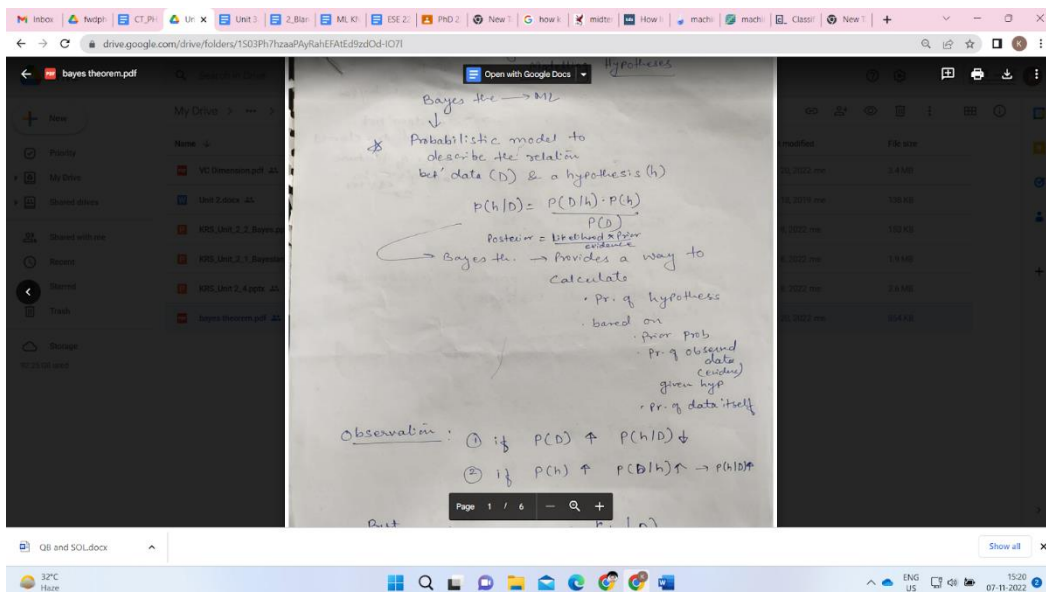
Unit 2

1	Comment whether k-NN lazy learner model. Justify your answer. Apply k-NN algorithm and predict the type of food type to which Tomato (sweet=6, crunch=4) belongs using minimum distance classifier with $k=1$. <table><tr><th>Ingredient</th><th>Sweet</th><th>Crunch</th><th>Food Type</th></tr><tr><td>Grape</td><td>8</td><td>5</td><td>Fruit</td></tr><tr><td>Green bean</td><td>3</td><td>7</td><td>Vegetable</td></tr><tr><td>Nuts</td><td>3</td><td>6</td><td>Protein</td></tr><tr><td>Orange</td><td>7</td><td>3</td><td>Fruit</td></tr></table>	Ingredient	Sweet	Crunch	Food Type	Grape	8	5	Fruit	Green bean	3	7	Vegetable	Nuts	3	6	Protein	Orange	7	3	Fruit
Ingredient	Sweet	Crunch	Food Type																		
Grape	8	5	Fruit																		
Green bean	3	7	Vegetable																		
Nuts	3	6	Protein																		
Orange	7	3	Fruit																		
	K-NN is a non parametric, which do not estimate any parameter. Explanation on this fact. $D(\text{Tomato, Grape})=\sqrt{(6-8)^2+(4-5)^2}=2.2$ $D(\text{Tomato, Green bean})=\sqrt{(6-3)^2+(4-7)^2}=4.2$ $D(\text{Tomato, Nuts})=\sqrt{(6-3)^2+(4-6)^2}=3.6$ $D(\text{Tomato, Orange})=\sqrt{(6-7)^2+(4-3)^2}=1.4$ Minimum distance between tomato and orange is 1.4,---- therefore tomato is classified as fruit.																				

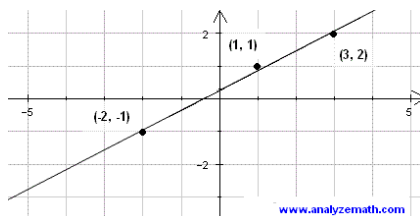
2	To build a classifier on discrete features using small training data, one will need to consider the scenario where some features have rare values that were never observed in the training data (e.g., the word 'Buenos Aires' does not appear in training for a text classification problem). To train a generalizable classifier, which classifier will you use among NB and LR.
	NB based on word frequency, and augment the frequency with pseudo-count, which corresponds to a Dirichlet prior over the multinomial frequency in a Bayesian setting.
3	To build a classifier on high-dimensional features using small training data, one will need to consider the scenario where many features are just irrelevant noises. To train a generalizable classifier, do you want to use NB or LR, how will you augment the original formulation of the classifier under a Bayesian or regularization setting?
	I would use LR, and use an L1 regularization as it encourages sparsity in the coefficients of the model.
4	Describe K-NN machine learning algorithm. How k will affect the K-NN model.
	KNN is one of several lazy learning algorithms that don't use a learning model to make predictions. By averaging the similarity between an incoming observation and the data already available, KNN creates predictions on the fly (just in time). KNN is a simple, supervised machine learning (ML) algorithm that can be used for classification or regression tasks and is also frequently used in missing value imputation. It is based on the idea that the observations closest to a given data point are the most "similar" observations in a data set, and we can therefore classify unforeseen points based on the values of the closest existing points. By choosing K, the user can select the number of nearby observations to use in the algorithm. K is the number of nearest neighbor to use. For classification, a majority vote is used to determine which class a new observation should fall into. Larger values of K are often more robust to outliers and produce more stable decision boundaries than very small values (K=3 would be better than K=1, which might produce undesirable results). If k is selected to be too low, the model becomes too specific and fails to generalize well. It also tends to be sensitive to noise. The model accomplishes a high accuracy on train set but will be a poor predictor on new, previously unseen data points. Therefore, we are likely to end up with an overfit model. If k is selected to be too large, the model becomes too generalized and fails to accurately predict the data points in both train and test sets. This situation is known as underfitting.

5	Compare and Contrast Parametric and Non-parametric techniques																																																											
	Parametric		Non-parametric																																																									
	Distribution based methods		Distribution free methods																																																									
	Data are generally unimodal		Real life problems are multimodal densities																																																									
	Type of density function such as Poisson, normal etc are known		Type of density function are unknown																																																									
	Estimation of parameters such as means, variance to make classification		Estimation of classification decision is based on without assuming forms of underlying densities.																																																									
	Fixed number of parameters		Arbitrary number of parameters																																																									
6	For the given dataset, design a Naive Bayes to classify a given test record X= (Refund =NO, Married, Income=120K) <table><tr><td>id</td><td>Refund</td><td>Marital Status</td><td>Taxable Income</td><td>Evade</td></tr><tr><td>1</td><td>Yes</td><td>Single</td><td>125K</td><td>NO</td></tr><tr><td>2</td><td>NO</td><td>Married</td><td>100K</td><td>NO</td></tr><tr><td>3</td><td>NO</td><td>Single</td><td>70K</td><td>NO</td></tr><tr><td>4</td><td>Yes</td><td>Married</td><td>120K</td><td>NO</td></tr><tr><td>5</td><td>NO</td><td>Divorced</td><td>95K</td><td>Yes</td></tr><tr><td>6</td><td>NO</td><td>Married</td><td>60K</td><td>NO</td></tr><tr><td>7</td><td>Yes</td><td>Divorced</td><td>220K</td><td>NO</td></tr><tr><td>8</td><td>NO</td><td>Single</td><td>85K</td><td>Yes</td></tr><tr><td>9</td><td>NO</td><td>Married</td><td>75K</td><td>NO</td></tr><tr><td>10</td><td>NO</td><td>Single</td><td>90K</td><td>Yes</td></tr></table>					id	Refund	Marital Status	Taxable Income	Evade	1	Yes	Single	125K	NO	2	NO	Married	100K	NO	3	NO	Single	70K	NO	4	Yes	Married	120K	NO	5	NO	Divorced	95K	Yes	6	NO	Married	60K	NO	7	Yes	Divorced	220K	NO	8	NO	Single	85K	Yes	9	NO	Married	75K	NO	10	NO	Single	90K	Yes
id	Refund	Marital Status	Taxable Income	Evade																																																								
1	Yes	Single	125K	NO																																																								
2	NO	Married	100K	NO																																																								
3	NO	Single	70K	NO																																																								
4	Yes	Married	120K	NO																																																								
5	NO	Divorced	95K	Yes																																																								
6	NO	Married	60K	NO																																																								
7	Yes	Divorced	220K	NO																																																								
8	NO	Single	85K	Yes																																																								
9	NO	Married	75K	NO																																																								
10	NO	Single	90K	Yes																																																								
	Given a Test Record:																																																											

	$X=(\text{Refund} = \text{NO}, \text{Married}, \text{Income}=120\text{K})$ $P(\text{Refund}=\text{Yes} \text{No}) = 3/7$ $P(\text{Refund}=\text{No} \text{No}) = 4/7$ $P(\text{Refund}=\text{Yes} \text{Yes}) = 0$ $P(\text{Refund}=\text{No} \text{Yes}) = 1$ $P(\text{Marital Status}=\text{Single} \text{No}) = 2/7$ $P(\text{Marital Status}=\text{Divorced} \text{No})=1/7$ $P(\text{Marital Status}=\text{Married} \text{No}) = 4/7$ $P(\text{Marital Status}=\text{Single} \text{Yes}) = 2/7$ $P(\text{Marital Status}=\text{Divorced} \text{Yes})=1/7$ $P(\text{Marital Status}=\text{Married} \text{Yes}) = 0$ $P(X \text{Class}=\text{No}) = P(\text{Refund}=\text{No} \text{Class}=\text{No}) \times P(\text{Married} \text{Class}=\text{No}) \times P(\text{Income}=120\text{K} \text{Class}=\text{No}) =$ $= \frac{4}{7} \times \frac{4}{7} \times 0.0072 = 0.0024$ $P(X \text{Class}=\text{Yes}) = P(\text{Refund}=\text{No} \text{Class}=\text{Yes}) \times P(\text{Married} \text{Class}=\text{Yes}) \times P(\text{Income}=120\text{K} \text{Class}=\text{Yes}) =$ $= 0 \times 1 \times 0 \times 1.2 \times 10^{-9} = 0$ Using Bayes Rule: Since $P(X \text{No}) P(\text{No}) > P(X \text{Yes})P(\text{Yes})$ Therefore $P(\text{No} X) > P(\text{Yes} X) \Rightarrow \text{Class} = \text{No}$ Ans: Evade (NO) =(Refund =NO, Married, Income=120K)
7	Discuss the Bayes Classifier. Show the steps for parameter estimation and analyze computational complexity.

	
8	Four Major issues to be considered in supervised learning The four major issue to be considered in supervised learning are as follows: 1. Bias/Variance trade off 2. Function complexity and amount of training data 3. Dimensionality of the input space 4. Noise in the input space
9	A sample from class A is located at $(x,y,z) = (1,2,3)$, a sample from class B is at $(7,4,5)$, and a sample from class C is at $(6,2,1)$. a) How would a sample at $(3,4,5)$ be classified using the single nearest neighbor technique and Euclidian Distance ? b) What if city block distances were used ? $D_A = \sqrt{((3-1))^2 + ((4-2))^2 + ((5-3))^2} = 3.46$ $D_B = \sqrt{((3-7))^2 + ((4-4))^2 + ((5-5))^2} = 4$ $D_C = \sqrt{((3-6))^2 + ((4-2))^2 + ((5-1))^2} = 5.38$ Hence, as D_A is minimum, $(3,4,5)$ will be classified as belonging to class A. City Block: $D_A = 3-1 + 4-2 + 5-3 = 6$ $D_B = 3-7 + 4-4 + 5-5 = 4$ $D_C = 3-6 + 4-2 + 5-1 = 9$ Hence, $(3,4,5)$ will be classified as belonging to class B.
10	Describe Shattering
	Shattering: H shatters S if $H[S] = 2^{ S }$

	A set of points S is shattered by H if there are hypothesis in H that splits S in all of the $2^{ S }$ possible ways, all possible ways of classifying points in S are achievable using concepts in H. Def: The VC Dimension of a hypothesis space H is the cardinality of the largest finite subset of set S that can be shattered by H. Dimensions of VC: If arbitrarily large finite sets can be shattered by H, then $VC(H)=\infty$ If there exists at least one subset of S of size d that can be shattered the $VC(H) \geq d$ If there exist $d+1$ point there is no set that can be shattered. If no subset of size d can be shattered, then $VC(H) < d$																																																							
11	Consider the data set shown in the table. <table><tr><th>Record</th><th>A</th><th>B</th><th>C</th><th>Class</th></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>+</td></tr><tr><td>2</td><td>0</td><td>0</td><td>1</td><td>-</td></tr><tr><td>3</td><td>0</td><td>1</td><td>1</td><td>-</td></tr><tr><td>4</td><td>0</td><td>1</td><td>1</td><td>-</td></tr><tr><td>5</td><td>0</td><td>0</td><td>1</td><td>+</td></tr><tr><td>6</td><td>1</td><td>0</td><td>1</td><td>+</td></tr><tr><td>7</td><td>1</td><td>0</td><td>1</td><td>-</td></tr><tr><td>8</td><td>1</td><td>0</td><td>1</td><td>-</td></tr><tr><td>9</td><td>1</td><td>1</td><td>1</td><td>+</td></tr><tr><td>10</td><td>1</td><td>0</td><td>1</td><td>+</td></tr></table> (a) Estimate the conditional probabilities using the m-estimate approach, with $p=1/2$ and $m=4$. (b) Use the estimate of conditional probabilities using the m-estimate approach to predict the class label for a test sample($A=0$, $B=1$, $C=0$) using.	Record	A	B	C	Class	1	0	0	0	+	2	0	0	1	-	3	0	1	1	-	4	0	1	1	-	5	0	0	1	+	6	1	0	1	+	7	1	0	1	-	8	1	0	1	-	9	1	1	1	+	10	1	0	1	+
Record	A	B	C	Class																																																				
1	0	0	0	+																																																				
2	0	0	1	-																																																				
3	0	1	1	-																																																				
4	0	1	1	-																																																				
5	0	0	1	+																																																				
6	1	0	1	+																																																				
7	1	0	1	-																																																				
8	1	0	1	-																																																				
9	1	1	1	+																																																				
10	1	0	1	+																																																				
	$P(k_i/C_j) = \frac{n_{ci} + m p_i}{n + m}$ (a) $P(A=0/+) = (2+2)/(5+4) = 4/9$ $P(A=0/-) = (3+2)/(5+4) = 5/9$ $P(B=1/+) = (1+2)/(5+4) = 3/9$ $P(B=1/-) = (2+2)/(5+4) = 4/9$ $P(C=0/+) = (1+2)/(5+4) = 3/9$																																																							

	$P(C=0/-) = (0+2)/(5+4) = 2/9$ (b) Let $P(A=0 , B=1 , C=0) = K$ $\Rightarrow P(A=0 , B=1 , C=0)$$= [P(A=0/+ , B=1/+ , C=0/+)*P(+)]/[P(A=0 , B=1 , C=0)]$$= [P(A=0/+) P(B=1/+) P(C=0/+) * P(+)] / K$$= [4/9 * 3/9 * 3/9 * 1/2] / K$$= 0.0247 / K$ $\Rightarrow P(-/A=0 , B=1 , C=0)$$= [P(A=0/- , B=1/- , C=0/-) * P(-)]/P(A=0 , B=1 , C=0)$$= [P(A=0/-) P(B=1/-) P(C=0/-) * P(-)] / K$$= [5/9 * 4/9 * 2/9 * 1/2] / K$$= 0.028 / K$ Therefore the Class Label should be (-).																
Unit 3																	
1	Briefly demonstrate the concept of Least Squares Regression. Consider the following set of points: $\{(-2 , -1) , (1 , 1) , (3 , 2)\}$ a) Find the least square regression line for the given data points. b) Plot the given points and the regression line in the same rectangular system of axes.																
	Let us organize the data in a table. <table><tr><th>x</th><th>y</th><th>x y</th><th>x²</th></tr><tr><td>-2</td><td>-1</td><td>-2</td><td>4</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td></tr><tr><td>3</td><td>2</td><td>6</td><td>9</td></tr></table> We now use the above formula to calculate a and b as follows 	x	y	x y	x ²	-2	-1	-2	4	1	1	1	1	3	2	6	9
x	y	x y	x ²														
-2	-1	-2	4														
1	1	1	1														
3	2	6	9														
2	The values of x and their corresponding values of y are shown in the table below <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>y</td><td>2</td><td>3</td><td>5</td><td>4</td><td>6</td></tr></table>	x	0	1	2	3	4	y	2	3	5	4	6				
x	0	1	2	3	4												
y	2	3	5	4	6												

	A] Find the least square regression line $y=ab + b$. B] Estimate the value of y when $x =10$.																																				
	A] <table><tr><td>X</td><td>Y</td><td>XY</td><td>X²</td></tr><tr><td>0</td><td>2</td><td>0</td><td>0</td></tr><tr><td>1</td><td>3</td><td>3</td><td>1</td></tr><tr><td>2</td><td>5</td><td>10</td><td>4</td></tr><tr><td>3</td><td>4</td><td>12</td><td>9</td></tr><tr><td>4</td><td>6</td><td>24</td><td>16</td></tr><tr><td>TOTAL=10</td><td>20</td><td>49</td><td>30</td></tr></table> Using least square regression formula $a = [n\sum xy - \sum x * \sum y] / [n\sum x^2 - (\sum x)^2]$ $= [(5 * 49) - (10*20)] / [5*30 -100]$ $= 0.9$ $b = 1/n [\sum y - a\sum x]$ $= 1/5 [20-9]$ $= 2.2$ Therefore, $Y = 0.9x + 2.2$ B] put $x=10$ $Y =0.9 * 10 + 2.2$ $Y = 11.2$	X	Y	XY	X ²	0	2	0	0	1	3	3	1	2	5	10	4	3	4	12	9	4	6	24	16	TOTAL=10	20	49	30								
X	Y	XY	X ²																																		
0	2	0	0																																		
1	3	3	1																																		
2	5	10	4																																		
3	4	12	9																																		
4	6	24	16																																		
TOTAL=10	20	49	30																																		
4	Consider the following set of points: $\{(-2 , -1) , (1 , 1) , (3 , 2)\}$ a) Find the least square regression line for the given data points. b) Plot the given points and the regression line in the same rectangular system of axes.																																				
	<table><tr><td>X</td><td>Y</td><td>$X - \bar{X}$</td><td>$Y - \bar{Y}$</td><td>$(X - \bar{X})^2$</td><td>$(\bar{X} - X)(Y - \bar{Y})$</td></tr><tr><td>-2</td><td>-1</td><td>-</td><td>-</td><td>7.1113</td><td>4.4446</td></tr><tr><td></td><td></td><td>2.6667</td><td>1.6667</td><td></td><td></td></tr><tr><td>1</td><td>1</td><td>0.3333</td><td>0.3333</td><td>0.1111</td><td>0.1111</td></tr><tr><td>3</td><td>2</td><td>2.3333</td><td>1.3333</td><td>5.4443</td><td>3.1110</td></tr><tr><td>$\Sigma=2$</td><td>$\Sigma=2$</td><td></td><td></td><td>$\Sigma=12.6667$</td><td>$\Sigma=7.6667$</td></tr></table>	X	Y	$X - \bar{X}$	$Y - \bar{Y}$	$(X - \bar{X})^2$	$(\bar{X} - X)(Y - \bar{Y})$	-2	-1	-	-	7.1113	4.4446			2.6667	1.6667			1	1	0.3333	0.3333	0.1111	0.1111	3	2	2.3333	1.3333	5.4443	3.1110	$\Sigma=2$	$\Sigma=2$			$\Sigma=12.6667$	$\Sigma=7.6667$
X	Y	$X - \bar{X}$	$Y - \bar{Y}$	$(X - \bar{X})^2$	$(\bar{X} - X)(Y - \bar{Y})$																																
-2	-1	-	-	7.1113	4.4446																																
		2.6667	1.6667																																		
1	1	0.3333	0.3333	0.1111	0.1111																																
3	2	2.3333	1.3333	5.4443	3.1110																																
$\Sigma=2$	$\Sigma=2$			$\Sigma=12.6667$	$\Sigma=7.6667$																																

$$\bar{X} = \bar{Y} = 2/3 = 0.6667$$

We know

$$Y = a + bx$$

$$b = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{\sum(X - \bar{X})^2} = 7.6667 / 12.6667 = 0.6053$$

$$a = \bar{y} - b\bar{x} = 2/3 - (0.6053)(2/3) = 0.2631$$

Therefore, $y = 0.2631 + 0.6053x$ is the least square regression line for given data points.

5

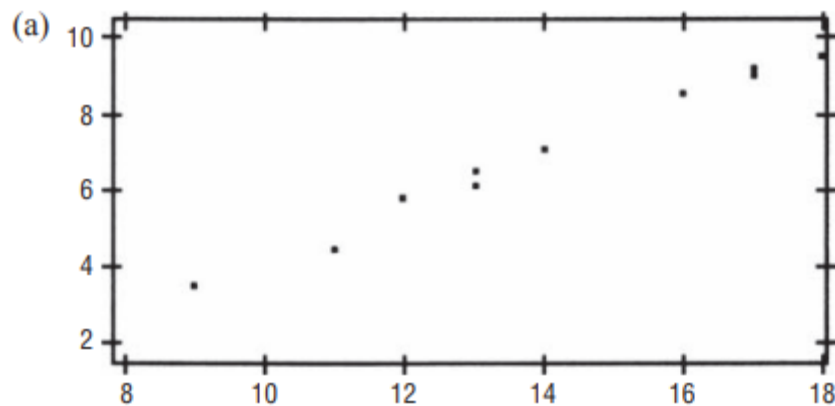
For the following set of data:

(a) Plot the scatter diagram.

(b) Develop the estimating equation that best describes the data.

(c) Predict Y for X = 10, 15, 20.

<i>X</i>	13	6	14	11	17	9	13	17	18	12
<i>Y</i>	6.2	8.6	7.2	4.5	9.0	3.5	6.5	9.3	9.5	5.7



(b)

<i>X</i>	<i>Y</i>	<i>XY</i>	<i>X</i> ²
13	6.2	80.6	169
16	8.6	137.6	256
14	7.2	100.8	196
11	4.5	49.5	121
17	9.0	153.0	289

X	Y	XY	X^2
9	3.5	31.5	81
13	6.5	84.5	169
17	9.3	158.1	289
18	9.5	171.0	324
12	5.7	68.4	144
$\Sigma X = 140$	$\Sigma Y = 70.0$	$\Sigma XY = 1,035.0$	$\Sigma X^2 = 2,038$

$$\bar{X} = 140 / 10 = 14 \quad \bar{Y} = 70.0 / 10 = 7.0$$

$$b = \frac{\Sigma XY - n\bar{X}\bar{Y}}{\Sigma X^2 - n\bar{X}^2} = \frac{1,035.0 - 10(14)(7.0)}{2,038 - 10(14)^2} = 0.7051$$

$$a = \bar{Y} - b\bar{X} = 7.0 - (0.7051)(14) = -2.8714$$

$$\text{Thus, } \hat{Y} = -2.8714 + 0.7051X.$$

Thus, $\hat{Y} = -2.8714 + 0.7051X$. If you used a computer regression package to do your computation, you probably got

$$\hat{Y} = -2.8718 + 0.7051X.$$

$$(c) \ X = 10, \hat{Y} = -2.8714 + 0.7051(10) = 4.1796$$

$$X = 15, \hat{Y} = -2.8714 + 0.7051(15) = 7.7051$$

$$X = 20, \hat{Y} = -2.8714 + 0.7051(20) = 11.2306$$

6

Cost accountants often estimate overhead based on the level of production. At the Standard Knitting Co., they have collected information on overhead expenses and units produced at different plants, and want to estimate a regression equation to predict future overhead

(a) Develop the regression equation for the cost accountants.

(b) Predict overhead when 50 units are produced.

(c) Calculate the standard error of estimate.

Overhead	191	170	272	155	280	173	234	116	153	178
Units	40	42	53	35	56	39	48	30	37	40

In this problem, Y = overhead and X = units produced

(a)

X	Y	XY	X^2	Y^2
40	191	7,640	1,600	36,481
42	170	7,140	1,764	28,900
53	272	14,416	2,809	73,984
35	155	5,425	1,225	24,025
56	280	15,680	3,136	78,400
39	173	6,747	1,521	29,929
48	234	11,232	2,304	54,756
30	116	3,480	900	13,456
37	153	5,661	1,369	23,409
40	178	7,120	1,600	31,684
$\Sigma X = 420$	$\Sigma Y = 1,922$	$\Sigma XY = 84,541$	$\Sigma X^2 = 18,228$	$\Sigma Y^2 = 395,024$

$$\bar{X} = \frac{420}{10} = 42 \quad \bar{Y} = \frac{1,922}{10} = 192.2$$

$$b = \frac{\Sigma XY - n\bar{X}\bar{Y}}{\Sigma X^2 - n\bar{X}^2} = \frac{84,541 - 10(42)(192.2)}{18,228 - 10(42)^2} = 6.4915$$

$$a = \bar{Y} - b\bar{X} = 192.2 - 6.4915(42) = -80.4430$$

$$\text{Thus, } \hat{Y} = -80.4430 + 6.4915X$$

$$(b) \hat{Y} = -80.4430 + 6.4915(50) = 244.1320$$

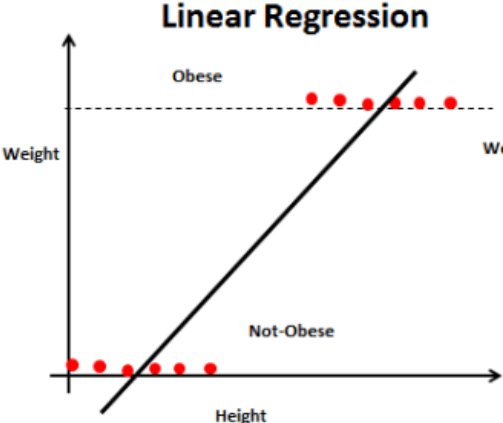
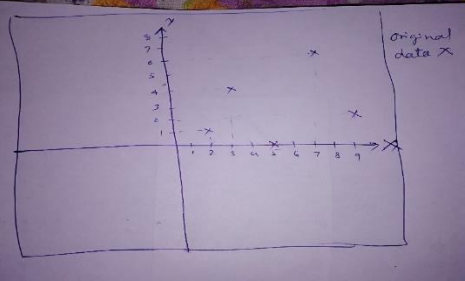
$$(c) s_e = \frac{\sqrt{\Sigma Y^2 - a\Sigma Y - b\Sigma XY}}{n-2}$$

$$= \sqrt{\frac{395,024 - (-80.4430)(1,922) - 6.4915(84,541)}{8}} = 10.2320$$

7

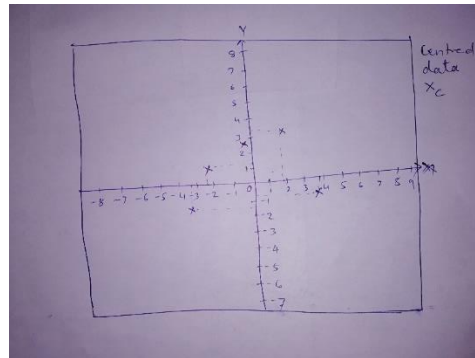
William C. Andrews, an organizational behavior consultant for Victory Motorcycles, has designed a test to show the company's supervisors the dangers of over supervising their workers. A worker from the assembly line is given a series of complicated tasks to perform. During the worker's performance, a supervisor constantly interrupts the worker to assist him or her in completing the tasks. The worker, upon completion of the tasks, is then given a psychological test designed to measure the worker's hostility toward authority (a high score equals low hostility). Eight different workers were assigned the tasks and then interrupted for the purpose of instructional assistance various numbers of times (line X). Their corresponding scores on the hostility test are revealed in line Y.

	X (number of times worker interrupted) 5 10 10 15 15 20 20 25 Y (worker's score on hostility test) 58 41 45 27 26 12 16 3 (a) Plot these data and identify the type of relationship among the datapoints. (b) Develop the equation that best describes the relationship between the number of times interrupted and the test score. (c) With the help of a developed relationship predict the expected test score if the worker is interrupted 18 times.
8	Distinguish between linear regression and logistic regression.
	The Similarities between Linear Regression and Logistic Regression Linear Regression and Logistic Regression both are supervised Machine Learning algorithms. Linear Regression and Logistic Regression, both the models are parametric regression i.e., both the models use linear equations for predictions That's all the similarities we have between these two models. However, functionality-wise these two are completely different. Following is the differences. The Differences between Linear Regression and Logistic Regression Linear Regression is used to handle regression problems whereas Logistic regression is used to handle the classification problems. Linear regression provides a continuous output, but Logistic regression provides discrete output. The purpose of Linear Regression is to find the best-fitted line while Logistic regression is one step ahead and fitting the line

	values to the sigmoid curve. The method for calculating loss function in linear regression is the mean squared error whereas for logistic regression it is maximum likelihood estimation. Linear Regression  Logistic Regression 
9	In simple linear regression, when β is not significantly different from zero we conclude that: a) X is a good predictor of Y b) there is no linear relationship between X and Y c) the relationship between X and Y is quadratic d) there is no relationship between X and Y
	b
Unit 4	
1	Given a matrix $X = \begin{bmatrix} 2 & 1 & 3 & 4 & 5 & 0 & 7 & 6 & 9 & 2 \end{bmatrix}$. Find the two principal components and that they are Orthogonal to each other
	 • Compute the mean of x and y. $\bar{x} = \frac{2 + 3 + 5 + 7 + 9}{5} = \frac{26}{5} = 5.2 \quad \bar{y} = \frac{1 + 4 + 0 + 6 + 2}{5} = \frac{13}{5} = 2.6$

$\bar{x}$	$\bar{y}$	$(x_i - \bar{x})$	$(y_i - \bar{y})$
5.2	2.6	-3.2	-1.6
5.2	2.6	-2.2	1.4
5.2	2.6	-0.2	-2.6
5.2	2.6	1.8	3.4
5.2	2.6	3.8	-0.6

Graph:



Subtracting the mean makes variance and covariance calculation easier by simplifying their equations. The variance and co-variance values are not affected by the mean value.

Compute the given table for further process:

$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$
5.12	10.24	2.56
3.08	4.84	1.96
0.52	0.04	6.76
6.12	3.24	11.56
2.28	14.44	0.36
6.25	32.8	23.2

- Calculate the covariance matrix $\text{Cov}(x, y)$ i.e., for two variables:

$$\text{Cov}(x, x) = \frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N-1} = 32.8/4 = 8.2$$

$$\text{Cov}(y, y) = \frac{\sum_{i=1}^N (y_i - \bar{y})^2}{N-1} = 23.2/4 = 5.6$$

$$\text{Cov}(x, y) = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{N-1} = 6.25/4 = 1.56$$

$$\text{Cov}(x, y) = S = \begin{bmatrix} 8.2 & 1.6 \\ 1.6 & 5.6 \end{bmatrix}$$

- Equation for eigenvalues and eigenvectors of covariance matrix S:

$$|(S - \lambda_k I)U_k| = 0$$

Using the equation compute λ_k where $k=2$ in our case λ_1 and λ_2

- Calculate the eigenvectors and eigenvalues of the covariance matrix:

$$\begin{vmatrix} 8.2 & 1.6 \\ 1.6 & 5.6 \end{vmatrix} - \lambda \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 0$$

$$\begin{vmatrix} 8.2 - \lambda & 1.6 \\ 1.6 & 5.6 - \lambda \end{vmatrix} = 0$$

$$(8.2 - \lambda)(5.6 - \lambda) - (1.6)^2 = 0$$

$$47.56 - 14\lambda + \lambda^2 - 2.56 = 0$$

$$\lambda^2 - 14\lambda + 45 = 0$$

$$(\lambda - 9)(\lambda - 5) = 0$$

$$\lambda_1 = 9$$

$$\lambda_2 = 5$$

$$\text{Eigen Values } \Lambda = \begin{bmatrix} 9 & 0 \\ 0 & 5 \end{bmatrix}$$

- Matrix of eigenvector

$$S\vec{v} = \Lambda\vec{v} \text{ -----eq. 1}$$

$$\vec{v} = \text{eigen vector}$$

$$\Lambda = \text{eigen values}$$

$$S = \text{Covariance matrix}$$

Let e_{11} and e_{12} are the two components of eigen vector v_1 of $\vec{v}$. Therefore, using the computed terms eq.1 can be written as:

- Considering $\lambda_1 = 9$

$$\begin{bmatrix} 8.2 & 1.6 \\ 1.6 & 5.6 \end{bmatrix} \begin{bmatrix} e_{11} \\ e_{12} \end{bmatrix} = 9 \begin{bmatrix} e_{11} \\ e_{12} \end{bmatrix}$$

$$8.2e_{11} + 1.6e_{12} = 9e_{11} \text{ ----1a}$$

$$1.6e_{11} + 5.6e_{12} = 9e_{12} \text{ ----2a}$$

From eq.1a

$$8.2e_{11} - 9e_{11} = -1.6e_{12}$$

$$-0.8e_{11} = -1.6e_{12}$$

$$e_{11} = 1.6/0.8 e_{12}$$

$$e_{11} = 2 e_{12} \text{ -----3a}$$

Eigenvector must be unit length, therefore $e_{11}^2 + e_{12}^2 = 1 \text{ -----4a}$

Using 3a and 4a

$$4 e_{12}^2 + e_{12}^2 = 1$$

$$e_{12}^2 = \frac{1}{5} = 0.2$$

$$e_{12} = \sqrt{0.2} = 0.447$$

Putting the value of e_{12} in eq .3a

$$e_{11} = 2e_{12} = 2 \times 0.447 = 0.894$$

$$e_{11} = 0.894, e_{12} = 0.447$$

- Considering $\lambda_2 = 5$

$$\begin{bmatrix} 8.2 & 1.6 \\ 1.6 & 5.6 \end{bmatrix} \begin{bmatrix} e_{21} \\ e_{22} \end{bmatrix} = 5 \begin{bmatrix} e_{21} \\ e_{22} \end{bmatrix}$$

$$8.2e_{21} + 1.6e_{22} = 5e_{21} \text{----1aa}$$

$$1.6e_{21} + 5.8e_{22} = 5e_{22} \text{----2aa}$$

From eq.1aa

$$8.2e_{21} - 5e_{21} = -1.6e_{22}$$

$$3.2e_{21} = -1.6e_{22}$$

$$e_{21} = -1.6/3.2 e_{22}$$

$$e_{21} = -0.5 e_{22} \text{-----3aa}$$

eigenvector must be unit length, therefore $e_{21}^2 + e_{22}^2 = 1$ -----4aa

Using 3aa and 4aa

$$0.25 e_{22}^2 + e_{22}^2 = 1$$

$$e_{22}^2 = \frac{1}{1.25} = 0.8$$

$$e_{22} = \sqrt{0.8} = 0.894$$

Putting the value of e_{22} in eq .3aa

$$e_{21} = -0.5e_{22} = 0.894 \times -0.5 = -0.447$$

$$e_{21} = -0.447, e_{22} = 0.894$$

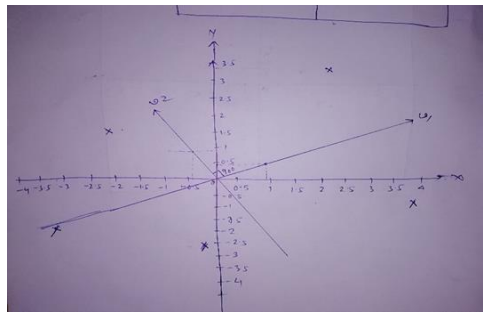
$$v_1 \quad v_2$$

$$\vec{v} = \begin{bmatrix} 0.894 & -0.447 \\ 0.447 & 0.894 \end{bmatrix}$$

$$(\lambda_1, v_1) = (9, \begin{matrix} 0.894 \\ 0.447 \end{matrix}) \text{ and } (\lambda_2, v_2) = (5, \begin{matrix} -0.447 \\ 0.894 \end{matrix})$$

Let's plot the eigenvector directions on the same graph. Eigenvectors are plotted as diagonal dotted lines on the plot. Note they are perpendicular to each other. Note one of the eigenvectors goes through the middle of the points, like drawing a line of best fit. The second eigenvector gives us the other, less important, pattern in the data, that all the points follow the main line, but are off to the side of the main line by some amount.

Graph



2

Use complete link agglomerative clustering to group the data described by the following distance matrix. Show the dendrograms.

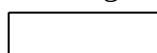
	A	B	C	D
A	0	1	4	5
B		0	2	6
C			0	3
D				0

Complete Link:

	A	B	C	D
A	0			
B	1	0		
C	4	2	0	
D	5	6	3	0

Minimum distance=1 for A, B. hence merge AB

Dendrogram



A

B

Distance



1

Find max distance from AB to other

$\max[\text{dist}\{A,B\},C]$

$= \max[\text{dist}(A,C),\text{dist}(B,C)]$

$= \max(4,2)$

$= 4$

$\max[\text{dist}\{(A,B),D\}]$

$= \max[\text{dist}(A,D),\text{dist}(B,D)]$

$= \max(5,6)$

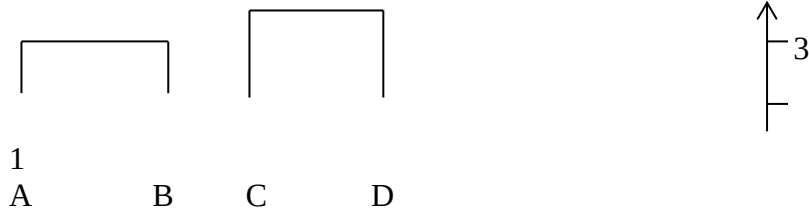
$= 6$

	AB	C	D
AB	1		
C	4	0	
D	6	3	0

Minimum distance=3 for C, D. Hence merge C, D.

Dendrogram

Distance



	AB	CD
AB	0	
CD	?	0

Find maximum distance from AB to CD

$\max[\text{dist}\{(A,B),(C,D)\}]$

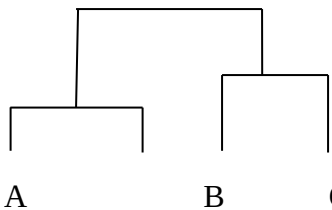
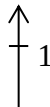
$= \max[\text{dist}(A,C),\text{dist}(A,D),\text{dist}(B,C),\text{dist}(B,D)]$

$= \max[4,5,2,6] = 6$

	AB	CD
AB	0	
CD	6	0

Merge AB and CD

	ABCD
ABCD	0

	Dendrogram  A B C D Distance 																																																		
3	Use single agglomerative clustering to group the data described by the following distance matrix. Show the dendrograms. <table><tr><td></td><td>A</td><td>B</td><td>C</td><td>D</td></tr><tr><td>A</td><td>0</td><td>1</td><td>4</td><td>5</td></tr><tr><td>B</td><td></td><td>0</td><td>2</td><td>6</td></tr><tr><td>C</td><td></td><td></td><td>0</td><td>3</td></tr><tr><td>D</td><td></td><td></td><td></td><td>0</td></tr></table>		A	B	C	D	A	0	1	4	5	B		0	2	6	C			0	3	D				0																									
	A	B	C	D																																															
A	0	1	4	5																																															
B		0	2	6																																															
C			0	3																																															
D				0																																															
	Single Link: <table><tr><td></td><td>A</td><td>B</td><td>C</td><td>D</td></tr><tr><td>A</td><td>0</td><td>1</td><td>4</td><td>5</td></tr><tr><td>B</td><td></td><td>0</td><td>2</td><td>6</td></tr><tr><td>C</td><td></td><td></td><td>0</td><td>3</td></tr><tr><td>D</td><td></td><td></td><td></td><td>0</td></tr></table>  <table><tr><td></td><td>A</td><td>B</td><td>C</td><td>D</td></tr><tr><td>A</td><td>0</td><td></td><td></td><td></td></tr><tr><td>B</td><td>1</td><td>0</td><td></td><td></td></tr><tr><td>C</td><td>4</td><td>2</td><td>0</td><td></td></tr><tr><td>D</td><td>5</td><td>6</td><td>3</td><td>0</td></tr></table> Min distance = 1 for A, B. Hence merge A, B at 1. Dendrogram  A B Distance  Find minimum distance from AB to other		A	B	C	D	A	0	1	4	5	B		0	2	6	C			0	3	D				0		A	B	C	D	A	0				B	1	0			C	4	2	0		D	5	6	3	0
	A	B	C	D																																															
A	0	1	4	5																																															
B		0	2	6																																															
C			0	3																																															
D				0																																															
	A	B	C	D																																															
A	0																																																		
B	1	0																																																	
C	4	2	0																																																
D	5	6	3	0																																															

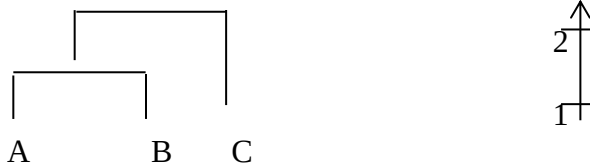
$\min[\text{dist}\{(A,B),C\}]$
 $= \min[\text{dist}(A,C),\text{dist}(B,C)]$
 $= \min[4,2]$
 $= 2$
 $\min[\text{dist}\{(A,B),D\}]$
 $= \min[\text{dist}(A,D),\text{dist}(B,D)]$
 $= \min[5,6]$
 $= 5$

	AB	C	D
AB	0		
C	2	0	
D	5	3	0

Minimum distance=2 for AB, C. Hence merge ABC.

Dendrogram

Distance



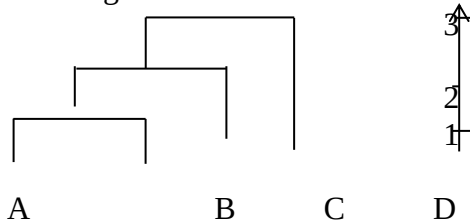
Find minimum distance from ABC to other

$\min[\text{dist}\{(AB,C),D\}]$
 $= \min[\text{dist}(AB,D),\text{dist}(C,D)]$
 $= \min[5,3]$
 $= 3$

	ABC	D
ABC	0	
D	3	0

Dendrogram

Distance



	ABCD
ABCD	0

4

Use the k-means algorithm and Euclidean distance to cluster the following 8 examples into 3 clusters:

A1=(2,10), A2=(2,5), A3=(8,4), A4=(5,8), A5=(7,5), A6=(6,4), A7=(1,2), A8=(4,9).

The distance matrix based on the Euclidean distance is given below:

	A1	A2	A3	A4	A5	A6	A7	A8
A1	0	25	36	13	50	52	65	5
A2		0	37	18	25	17	10	20
A3			0	25	2	2	53	41
A4				0	13	17	52	2
A5					0	2	45	25
A6						0	29	29
A7							0	58
A8								0

Suppose that the initial seeds (centres of each cluster) are A1, A4 and A7. Apply the k-means algorithm for 1 epoch only. At the end of 1st epoch show:

- The new clusters (i.e., the examples belonging to each cluster)
- The centres of the new clusters
- Draw a 10 by 10 space with all the 8 points and show the clusters after the first epoch and the new centroids.
- How many more iterations are needed to converge? Draw the result for each epoch.

a)

d(a,b) denotes the Euclidean distance between a and b. It is obtained directly from the distance matrix or calculated as follows: $d(a,b) = \sqrt{(x_b - x_a)^2 + (y_b - y_a)^2}$

seed1=A1=(2,10), seed2=A4=(5,8), seed3=A7=(1,2)

epoch1 – start:

A1:

d(A1, seed1)=0 as A1 is seed1

d(A1, seed2)= 13 >0

d(A1, seed3)= 65 >0

- A1 ∈ cluster1

A2:

d(A2,seed1)= 25 = 5

d(A2, seed2)= 18 = 4.24

d(A2, seed3)= 10 = 3.16 □ smaller

- A2 ∈ cluster3

A3:

d(A3, seed1)= 36 = 6

d(A3, seed2)= 25 = 5 □ smaller

d(A3, seed3)= 53 = 7.28

- A3 ∈ cluster2

A4:

d(A4, seed1)= 13

$d(A4, \text{seed2})=0$ as A4 is seed2

$d(A4, \text{seed3})= 52 >0$

- $A4 \in \text{cluster2}$

A5:

$d(A5, \text{seed1})= 50 = 7.07$

A6:

$d(A6, \text{seed1})= 52 = 7.21$

$d(A5, \text{seed2})= 13 = 3.60$ □ smaller

$d(A5, \text{seed3})= 45 = 6.70$

- $A5 \in \text{cluster2}$

$d(A6, \text{seed2})= 17 = 4.12$ □ smaller

$d(A6, \text{seed3})= 29 = 5.38$

- $A6 \in \text{cluster2}$

A7:

$d(A7, \text{seed1})= 65 >0$

$d(A7, \text{seed2})= 52 >0$

$d(A7, \text{seed3})=0$ as A7 is seed3

- $A7 \in \text{cluster3}$

A8:

$d(A8, \text{seed1})= 5$

$d(A8, \text{seed2})= 2$ □ smaller

$d(A8, \text{seed3})= 58$

- $A8 \in \text{cluster2}$

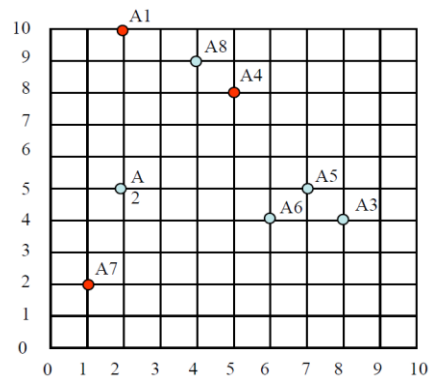
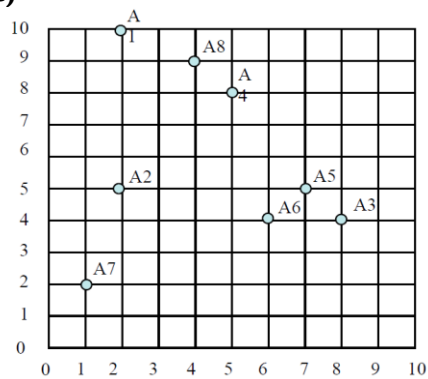
end of epoch1

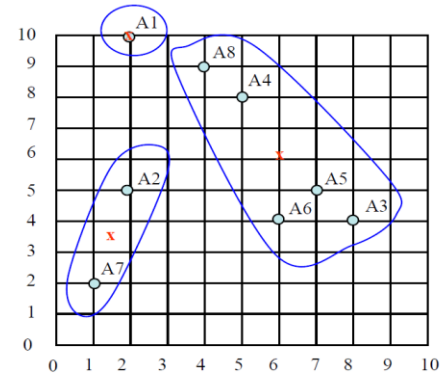
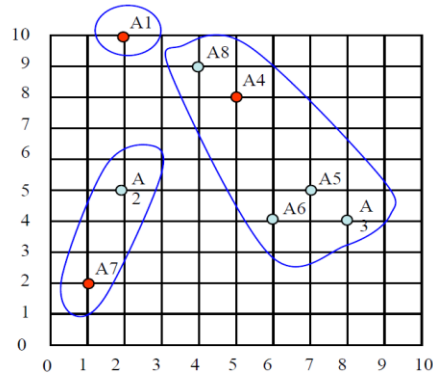
new clusters: 1: {A1}, 2: {A3, A4, A5, A6, A8}, 3: {A2, A7}

b) centers of the new clusters:

$C1= (2, 10)$, $C2= ((8+5+7+6+4)/5, (4+8+5+4+9)/5) = (6, 6)$, $C3= ((2+1)/2, (5+2)/2) = (1.5, 3.5)$

c)





d)

We would need two more epochs. After the 2nd epoch the results would be:

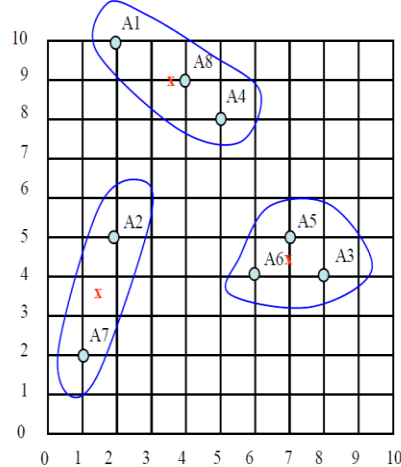
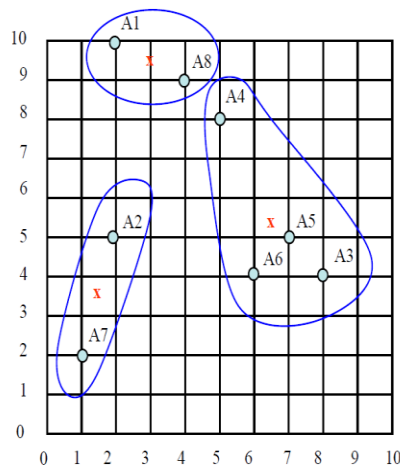
1: {A1, A8}, 2: {A3, A4, A5, A6}, 3: {A2, A7}

with centers $C1=(3, 9.5)$, $C2=(6.5, 5.25)$ and $C3=(1.5, 3.5)$.

After the 3rd epoch, the results would be:

1: {A1, A4, A8}, 2: {A3, A5, A6}, 3: {A2, A7}

with centers $C1=(3.66, 9)$, $C2=(7, 4.33)$ and $C3=(1.5, 3.5)$.



5

Describe feature selection process. With a neat diagram discuss filter and wrapper approach of feature selection.

Process to find a subset of the original set of variables, or features, to get a smaller subset which can be used to model the problem.

For example, consider $F=\{x_1, x_2, \dots, x_N\}$ as a given set of features initially. The subset of F can be $F' \subset F = \{x_1', x_2' \dots x_M'\}$, where $M < N$.

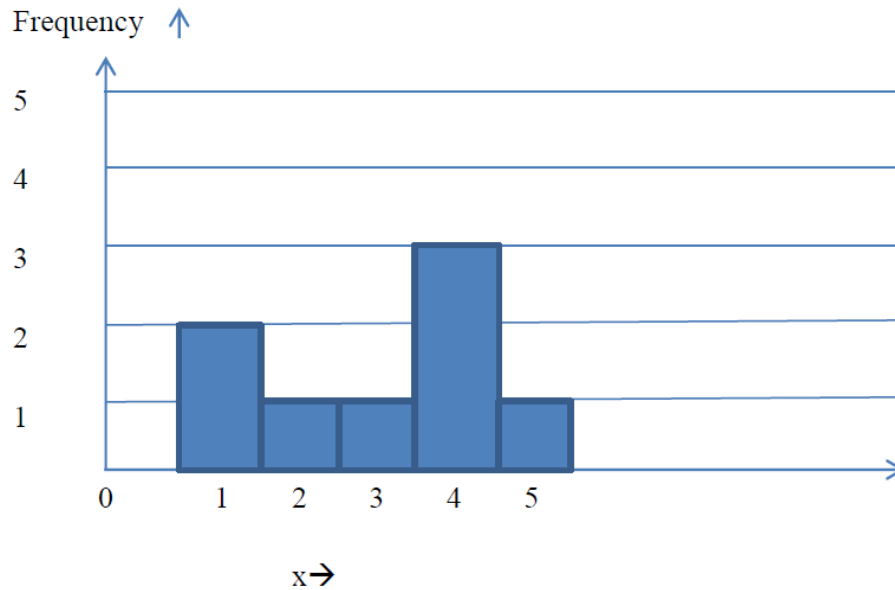
The selected subset optimizes certain criteria's such as, it either improves or maintains the accuracy, simplify classifier complexity, and to avoid over fitting the data in order to make further analysis possible.

For given N features/dimension there could be 2^N possible subset. For example

	$N=3$, there could be 8 possible subsets. Feature selection can be done by two ways: Filter: The filter method extract features from the data without any learning involved. Wrapper: The wrapper method use learning techniques to evaluate which features are useful. Filter Method It is an unsupervised method where we don't evaluate the subset over training set. We evaluate it on information content in an unsupervised way. Filter methods select features based on a performance measure regardless of the employed data modelling algorithm. Only after the best features are found, the modelling algorithms can use them. Generally, filter method Look at input only Select the subset that has the most information Is a one-shot process (not iterative). Provides a set of "the most important" variables as the resulting subset independently of the employed model Wrapper method It is a supervised method that evaluate over training examples using some learning algorithms. Generally, wrapper methods Train using selected subset Estimate error on validation set Selects a subset of variables taking into account the model that will use them It is an iterative process In each iteration, several subsets of input variables is generated and tested on the particular model type According to the success of the model for individual feature subsets, it is chosen which subsets will be tested in the next iteration. Feature selection is here part of the model training; we need separate testing data to evaluate the final model error. Thus, for classification tasks, a wrapper will evaluate subsets based on the classifier performance (e.g. Naïve Bayes or SVM), while for clustering, a wrapper will evaluate subsets based on the performance of a clustering algorithm (e.g. K-means). The evaluation is repeated for each subset, and the subset generation is dependent on the search strategy, in the same way as with filters. Wrappers are much slower than filters in finding sufficiently good subsets because they depend on the resource demands of the modelling algorithm.
--	---

6	With respect to clustering answer each of the following. - Define Hard and Soft clustering - Define non-hierarchical and hierarchical clustering - Three convergence criteria of clustering. - List and define two internal measures of clusters. - State and brief about any two weakness of k-means algorithm.
	Hard clustering: Each data point x_i belongs to exactly one cluster Soft clustering: A data point x_i can belong to more than one cluster with a probability $p_i \in [0,1]$ A non-hierarchical method generates a classification by partitioning a dataset, giving a set of (generally) non-overlapping groups having no hierarchical relationships between them. Hierarchical clustering is characterized by the development of a hierarchy or tree-like structure. no (or minimum) re-assignments of data points to different clusters, no (or minimum) change of centroids, or minimum decrease in the sum of squared error (SSE), Cluster Cohesion: Measures how closely related are objects in a cluster Example: SSE Cluster Separation: Measure how distinct or well-separated a cluster is from other clusters K-means has problems when clusters are of differing - Sizes - Densities - Non-globular shapes K-means has problems when the data contains outliers.
7	Define hierarchical clustering. With respect to hierarchical clustering, discuss agglomerative and divisive in terms of definition, diagrammatic representation, and algorithm.
	Hierarchical clustering is characterized by the development of a hierarchy or tree-like structure. Agglomerative clustering starts with each object in a separate cluster. Clusters are formed by grouping objects into bigger and bigger clusters. Agglomerative (bottom up) clustering.

	Initialization: Each object is a cluster Iteration: Merge two clusters which are most like each other. Until all objects are merged into a single cluster Agglomerative methods are commonly used in marketing research. They consist of linkage methods, variance methods, and centroid methods. Divisive clustering starts with all the objects grouped in a single cluster. Clusters are divided or split until each object is in a separate cluster. Initialization: All objects stay in one cluster Iteration: Select a cluster and split it into two sub clusters Until each leaf cluster contains only one object
8	Differentiate between partitioning clustering and hierarchical clustering.
9	You are given a dataset $D = \{0, 1, 1, 2, 3, 4, 4, 4, 5\}$. Using the technique from non-parametric density estimation answer the following questions. a) Draw a histogram of D with a bin width of 1 and bins centred at $\{0, 1, 2, 3, 4, 5\}$. b) Write the formula for the Kernel density estimate given an arbitrary Kernel. c) Select a triangle Kernel as your window function.
	The histogram of D is as follows:



The formula for the Kernel density estimate is as follows:-
 $f(k) = 1 / qh * \text{summation of } \{w(i) * K [x * X(i) / h]\}$ from $i=1$ to n
 where $q = \text{summation of } W(i)$, if weights are frequency weights or analytic weights and $q = 1$, if weights are importance weights. If weights are not used then $W(i) = 1$, for $i = 1$ to n . K-density includes seven different kernel functions. The Epanechnikov is the default function if no other Kernel is specified and is the most efficient in minimizing the mean integrated squared error.

For the triangular kernel,
 $K[z] = 1 - |z|$, if $|z| < 1$
 $= 0$, otherwise

Where h is called as the window width or band width. If the window width is not specified, it is determined as,
 $m = \min \{ \text{sqrt of } [\text{variance}(x)], \text{interquartile range}(x) / 1.349 \}$
 $h = 0.9m / \text{fifth root of } n$
 where x is the variable for which we wish to estimate the kernel and n is the number of observations.

Unit 5

1

ISRO wants to discriminate between Martians (M) and Humans (H) based on the following features: Green E {N,Y}, Legs E {2,3}, Height E {S,T}, Smelly E {N,Y}. The training data is as follows:

a. Which attribute will be the root of the decision tree:

Species	Green	Legs	Height	Smelly
---------	-------	------	--------	--------

M	N	3	S	Y
M	Y	2	T	N
M	Y	3	T	N
M	N	2	S	Y
M	Y	3	T	N
H	N	2	T	Y
H	N	2	S	N
H	N	2	T	N
H	Y	2	S	N
H	N	2	T	Y

Page No. 1
Date: / /

Q.3-

$$\begin{array}{c}
 10 \\
 \swarrow \quad \searrow \\
 5M \quad 5H
 \end{array}$$

$$F(S) = E(5M, 5H) = 1$$

$$\begin{array}{c}
 \text{Green}(10) \\
 \swarrow \quad \searrow \\
 N(6) \quad Y(4) \\
 [2M, 4H] \quad [3M, 1H]
 \end{array}$$

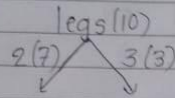
$$\text{Gain}(S, \text{Green}) = E(S) - \sum_{V \in \text{values}(A)} \frac{|S_v|}{S} E(S_v)$$

$$\begin{aligned}
 E(S_N) &= E(2M, 4H) \\
 &= -\frac{2}{6} \log_2 \left(\frac{2}{6} \right) - \frac{4}{6} \log_2 \left(\frac{4}{6} \right) \\
 &= 0.5283 + 0.3899 \\
 &= 0.9182
 \end{aligned}$$

$$\begin{aligned}
 E(S_Y) &= E(3M, 1H) \\
 &= -\frac{3}{4} \log_2 \left(\frac{3}{4} \right) - \frac{1}{4} \log_2 \left(\frac{1}{4} \right) \\
 &= 0.3113 + 0.5 \\
 &= 0.8113
 \end{aligned}$$

$$\begin{aligned}
 \text{Gain}(S, \text{Green}) &= 1 - \left[\frac{6}{10} \times 0.9182 + \frac{4}{10} \times 0.8113 \right] \\
 &= 1 - [0.5509 + 0.3245] \\
 \text{Gain}(S, \text{Green}) &= 0.1246
 \end{aligned}$$

$$\text{Gain}(S, \text{legs}) = E_S - \sum_{v \in \text{values}(A)} \frac{|S_v|}{S} E(S_v)$$



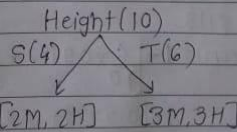
[2M, 5H] [3M, 0H]

$$\begin{aligned} E(S_2) &= E(2M, 5H) \\ &= -\frac{2}{7} \log_2 \left(\frac{2}{7} \right) - \frac{5}{7} \log_2 \left(\frac{5}{7} \right) \\ &= 0.5163 + 0.3467 \\ &= 0.863 \end{aligned}$$

$$\begin{aligned} E(S_3) &= E(3M, 0H) \\ &= -\frac{3}{3} \log_2 \left(\frac{3}{3} \right) - 0 \\ &= 0 \end{aligned}$$

$$\begin{aligned} \text{Gain}(S, \text{legs}) &= 1 - \left[\frac{7}{10} \times 0.863 + \frac{3}{10} \times 0 \right] \\ &= 1 - 0.6041 \end{aligned}$$

$$\text{Gain}(S, \text{legs}) = 0.3959$$



$$E(S_s) = E(2M, 2H)$$

$$= 1$$

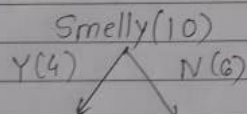
$$E(S_r) = E(3M, 3H)$$

$$= 1$$

$$\text{Gain}(S, \text{Height}) = 1 - \left[\frac{4}{10} \times 1 + \frac{6}{10} \times 1 \right]$$

$$= 1 - 1$$

$$\boxed{\text{Gain}(S, \text{Height}) = 0}$$



$$E(S_Y) = E(2M, 2H)$$

$$= 1$$

$$E(S_N) = E(3M, 3H)$$

$$= 1$$

$$\text{Gain}(S, \text{Smelly}) = 1 - \left[\frac{4}{10} \times 1 + \frac{6}{10} \times 1 \right]$$

$$= 1 - 1$$

$$\boxed{\text{Gain}(S, \text{Smelly}) = 0}$$

Gain(S, legs) is greater

∴ The required attribute would be legs for the root of the decision tree.

2

Assn 8. Given the training data (*Buy Computer* data), build a decision tree and predict the class of the following new example: age $\leq$ 30, income=medium, student=yes, credit-rating=fair.

RID	age	income	student	credit_rating	Class: buys_computer
1	≤ 30	high	no	fair	no
2	≤ 30	high	no	excellent	no
3	31 ... 40	high	no	fair	yes
4	> 40	medium	no	fair	yes
5	> 40	low	yes	fair	yes
6	> 40	low	yes	excellent	no
7	31 ... 40	low	yes	excellent	yes
8	≤ 30	medium	no	fair	no
9	≤ 30	low	yes	fair	yes
10	> 40	medium	yes	fair	yes
11	≤ 30	medium	yes	excellent	yes
12	31 ... 40	medium	no	excellent	yes
13	31 ... 40	high	yes	fair	yes
14	> 40	medium	no	excellent	no

$$\text{Entropy}(s) = [9\text{yes}, 5\text{no}]$$

$$= -\frac{9}{14}\log\left(\frac{9}{14}\right) - \frac{5}{14}\log\left(\frac{5}{14}\right)$$

$$= -0.9402$$

$$1.] \text{ Gain}(S, \text{Age}) = E(s) - \sum_{v \in \text{values}} \left(\frac{|S_v|}{S}\right) \text{Ent}(S_v)$$

$$= 0.9402 - \left[\frac{5}{14} \left(-\frac{2}{14}\log\frac{2}{14} - \frac{3}{14}\log\frac{3}{14}\right)\right] + \left[-\frac{4}{14} \left(-\frac{4}{4}\log\frac{4}{4}\right)\right] + \left[\frac{5}{14} \left(-\frac{3}{5}\log\frac{2}{5} - \frac{2}{5}\log\frac{2}{5}\right)\right]$$

$$= 0.9402 - \left[\left(\frac{5}{14} \times 0.9709\right) + 0 + \left(\frac{5}{14} \times 0.9709\right)\right]$$

$$= 0.9402 - 0.6935$$

$$= 0.2467$$

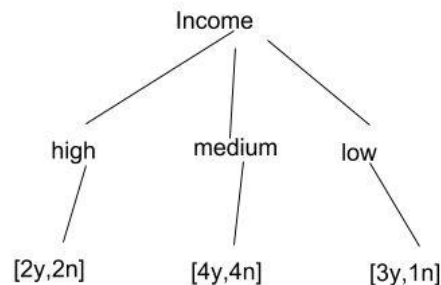


Diagram2

$$2.] \text{ Gain}(S, \text{Income}) = E(s) - \sum_{v \in \text{values}} \left(\frac{|S_v|}{S}\right) \text{Ent}(S_v)$$

$$= 0.9402 - \left[\frac{4}{14} \left(-\frac{2}{4}\log\frac{2}{4} - \frac{2}{4}\log\frac{2}{4}\right)\right] + \left[\frac{6}{14} \left(-\frac{4}{6}\log\frac{4}{6} - \frac{2}{6}\log\frac{2}{6}\right)\right] + \left[\frac{4}{14} \left(-\frac{3}{4}\log\frac{3}{4} - \frac{1}{4}\log\frac{1}{4}\right)\right]$$

$$= 0.9402 - \left[\left(\frac{4}{14} \times 1\right) + \left(\frac{6}{14} \times 0.9182\right) + \left(\frac{4}{14} \times 0.810\right)\right]$$

$$=0.9402-0.911$$

$$=0.0292$$

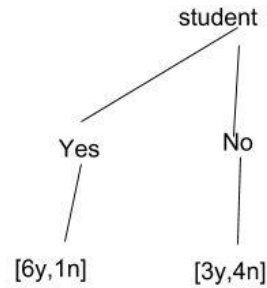


Diagram3

$$3.] \text{ Gain}(S, \text{Student}) = E(s) - \sum_{v \in \text{values}} \left(\frac{|S_v|}{S} \right) \text{Ent}(S_v)$$

$$=0.9402 - \left[\frac{7}{14} \left(-\frac{6}{7} \log \frac{6}{7} - \frac{1}{7} \log \frac{1}{7} \right) \right] + \left[\frac{7}{14} \left(-\frac{3}{7} \log \frac{3}{7} - \frac{4}{7} \log \frac{4}{7} \right) \right]$$

$$=0.9402 - \left[\left(\frac{7}{14} \times 0.5916 + \frac{7}{14} \times 0.9852 \right) \right]$$

$$=0.9402 - 0.7834$$

$$=0.1518$$

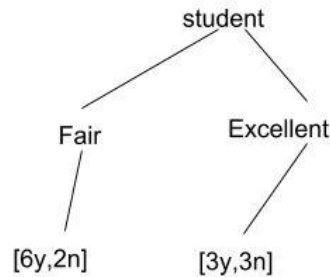


Diagram4

$$4.] \text{ Gain}(S, \text{Credit_ranking}) = E(s) - \sum_{v \in \text{values}} \left(\frac{|S_v|}{S} \right) \text{Ent}(S_v)$$

$$=0.9402 - \left[\frac{8}{14} \left(-\frac{6}{8} \log \frac{6}{8} - \frac{2}{8} \log \frac{2}{8} \right) \right] + \left[\frac{6}{14} \times 1 \right]$$

$$=0.9402 - \left[\left(\frac{8}{14} \times 0.8112 + \frac{6}{14} \right) \right]$$

$$=0.9402 - 0.8921$$

$$=0.0481$$

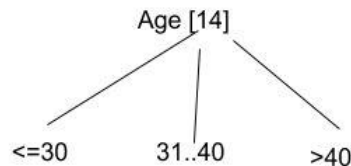


Diagram 5

For <=30 we have three attributes – income, student, credit_rating

$$E(<=30) = -\frac{2}{5} \log \left(\frac{2}{5} \right) - \frac{3}{5} \log \left(\frac{3}{5} \right)$$

$$=0.9709$$

$$\begin{aligned}
 &\text{Gain}(\text{Income}) \\
 &= E(<=30) - \left[\frac{2}{5}(0,2) + \frac{2}{5}(1,1) + \frac{1}{5}(1,0) \right] \\
 &= 0.9709 - 0.4 \\
 &= 0.5709
 \end{aligned}$$

$$\begin{aligned}
 &\text{Gain}(\text{Student}) \\
 &= 0.9709 - \left[\frac{2}{5}(2,0) + \frac{2}{5}(1,1) + \frac{3}{5}(0,3) \right] \\
 &= 0.9709
 \end{aligned}$$

$$\begin{aligned}
 &\text{Gain}(\text{credit_rating}) \\
 &= 0.9709 - \left[\frac{3}{5}(-1,2) + \frac{2}{5}(1,1) \right] \\
 &= 0.9709 - \left[\frac{3}{5} \left(-\frac{1}{3} \log\left(\frac{2}{3}\right) - \frac{2}{3} \log\left(\frac{2}{3}\right) \right) \right] \\
 &= 0.9709 - \left[\frac{3}{5} \times 0.9182 + \frac{2}{5} \right] \\
 &= 0.9709 - 0.9509 \\
 &= 0.02
 \end{aligned}$$

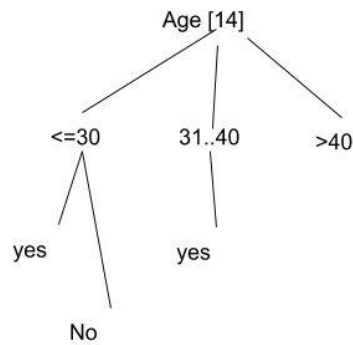


Diagram6

$$\begin{aligned}
 &\text{Now, for } >40 \\
 &E(>40) = 0.9709 \\
 &= 0.9709 - \left[\frac{3}{5}(2,1) + \frac{2}{5}(1,1) \right] \\
 &= 0.9709 - \left[\frac{3}{5} \left(-\frac{2}{3} \log\left(\frac{2}{3}\right) - \frac{1}{3} \log\left(\frac{1}{3}\right) \right) + \frac{2}{5} \right] \\
 &= 0.02
 \end{aligned}$$

$$\begin{aligned}
 &\text{Gain}(\text{Student}) \\
 &= 0.9709 - \left[\frac{3}{5}(2,1) + \frac{2}{5}(1,1) + \frac{2}{5}(1,1) \right] \\
 &= 0.2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Gain}(\text{credit_rating}) \\
 &= 0.9709 - \left[\frac{3}{5}(3,0) + \frac{2}{5}(0,2) \right] \\
 &= 0.9709 - [0+0] \\
 &= 0.9709
 \end{aligned}$$

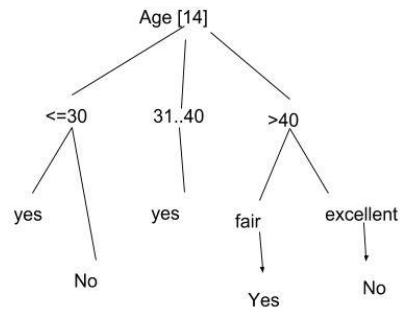


Diagram7

For the new examples

< age<=30,income=medium,student=yes,credit_rating=fair >

If age<=30 and student=yes, then buy_comuter

3

DAY	OUTLOOK	TEMPERATURE	HUMIDITY	WIND	PLAY TENNIS
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Weak	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Strong	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Develop a classification model using decision tree ID3 for the data in the above table. For the developed model obtain the play prediction for the day <sunny, cool, high, strong>.

Four attributes used for classification:

Outlook = { Sunny, Overcast, Rainy }

Temperature = { Hot, Mild, Cold }

Humidity = { High, Normal }

Wind = { Weak, Strong }

Target Attribute:

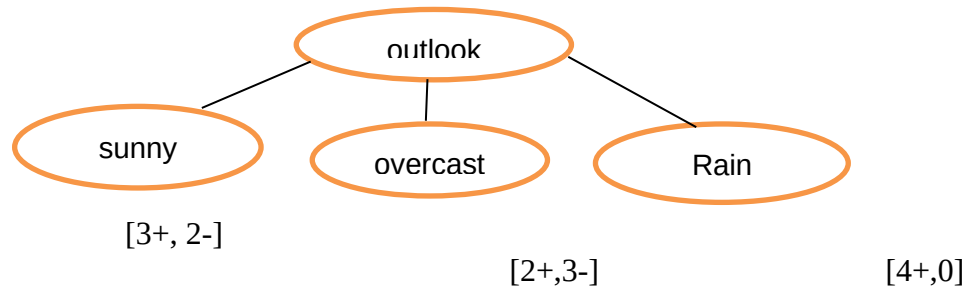
Playtennis = { Yes, No }

Given 14 training examples:

9 positive

5 negative

Applying ID3 algorithm-



$$\text{Entropy } [9+, 5-] = (-9/14)\log_2(9/14) - 5/14\log_2(5/14) = 0.940$$

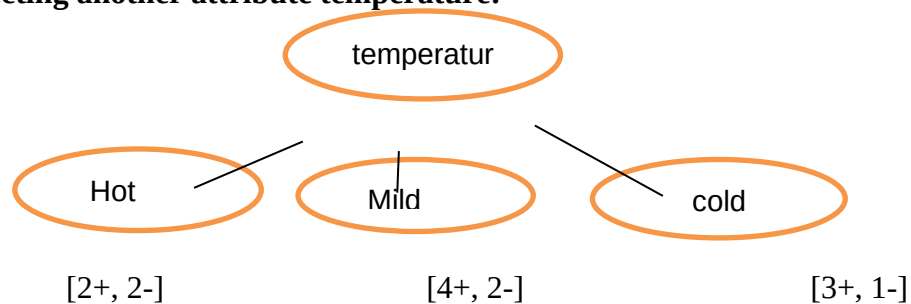
$$E(\text{sunny}) = E[2+, 3-] = -2/5\log_2(2/5) - 3/5\log_2(3/5) = 0.971$$

$$E(\text{overcast}) = E[4+, 0] = 0$$

$$E(\text{rain}) = E[3+, 2-] = -3/5\log_2(3/5) - 2/5\log_2(2/5) = 0.971$$

$$\text{Gain}(S, \text{Outlook}) = 0.940 - (5/14)*0.971 - (4/14)*0 - (5/14)*0.971 = 0.247$$

Selecting another attribute temperature:



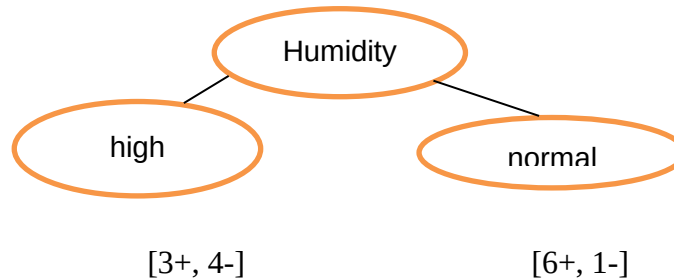
$$E(\text{Hot}) = E[2+, 2-] = 1$$

$$E(\text{Mild}) = E[4+, 2-] = -4/6\log_2(4/6) - 2/6\log_2(2/6) = 0.911$$

$$E(\text{cold}) = E[3+, 1-] = 3/4\log_2(3/4) - 1/4\log_2(1/4) = 0.811$$

$$\text{Gain}(S, \text{Temperature}) = 0.940 - (4/14)*1 - (6/14)*0.911 - (4/14)*0.811 \\ = 0.029$$

Selecting another attribute humidity:

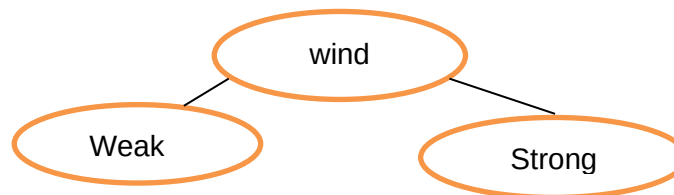


$$E(\text{high}) = E[3+, 4-] = -3/7\log_2(3/7) - 4/7\log_2(4/7) \\ = 0.985$$

$$E(\text{normal}) = E[6+, 1-] = -6/7\log_2(6/7) - 1/7\log_2(1/7) \\ = 0.592$$

$$\text{Gain}(S, \text{Humidity}) = 0.940 - (7/14)*0.985 - (7/14)*0.592 \\ = 0.151$$

Selecting another attribute wind:



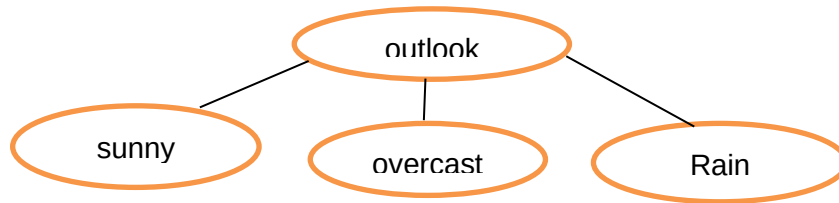
$$E(\text{weak}) = E[6+, 2-] = -6/7\log_2(6/7) - 2/7\log_2(2/7) \\ = 0.811$$

$$E(\text{strong}) = E[3+, 3-] = 1$$

$$\text{Gain}(S, \text{wind}) = 0.940 - (8/14)*0.811 - (6/14)*1 \\ = 0.048$$

Now, since outlook has highest gain, therefore, it is selected as root:

[9+, 5-]



? yes _{+, 0]} ? _{-, 2-]}

$$\text{Gain} (S_{\text{sunny}}, \text{Humidity}) = 0.970 - (3/5)*0 - (2/5)*0 = 0.970$$

$$\text{Gain} (S_{\text{sunny}}, \text{temp}) = 0.970 - (2/5)*0 - (2/5)*1 - (1/5)*0 = 0.570$$

$$\text{Gain} (S_{\text{sunny}}, \text{wind}) = 0.970 - (2/5)*1 - (3/5)*0.918 = 0.019$$

Since, gain of humidity is highest, therefore it is selected as next best attribute.

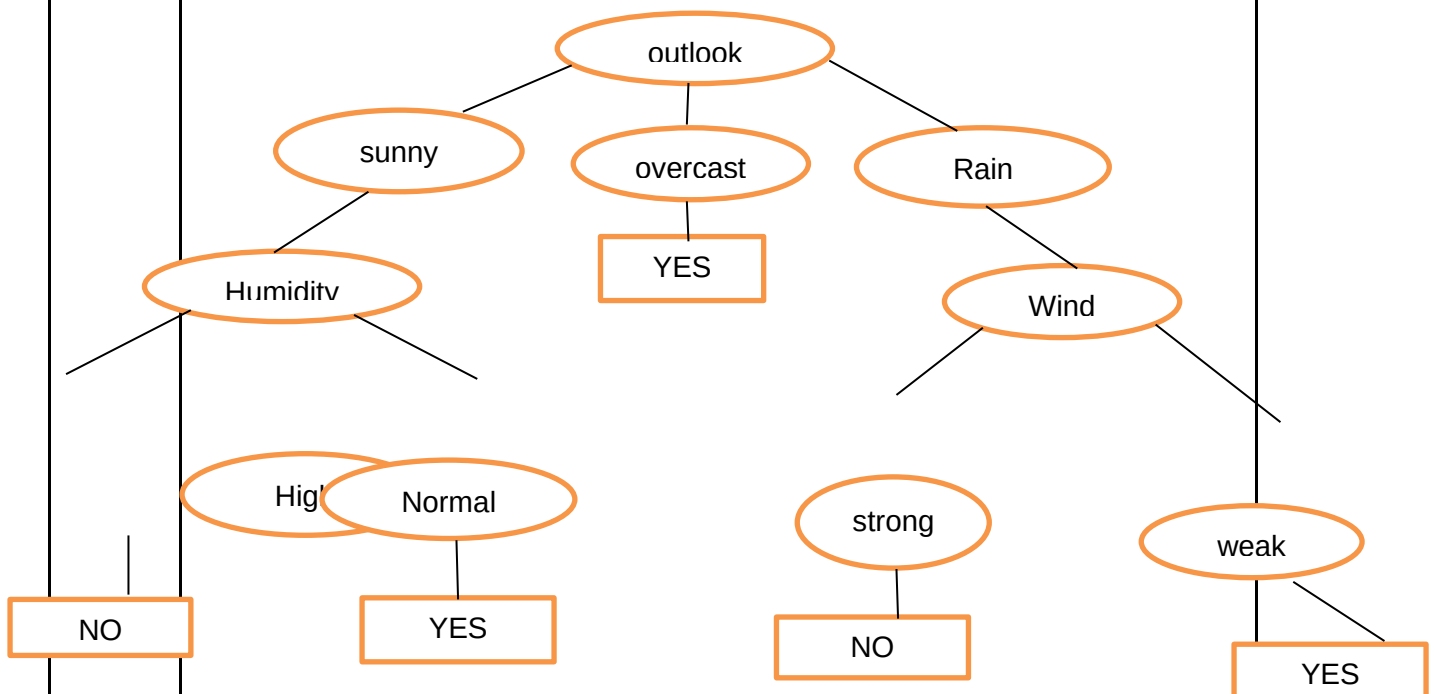


Figure: Final decision tree

Now, we can make if-then rules from the above decision tree as follows:

If (Outlook = Sunny)

If (Humidity = High) then playtennis = NO

else playtennis = YES

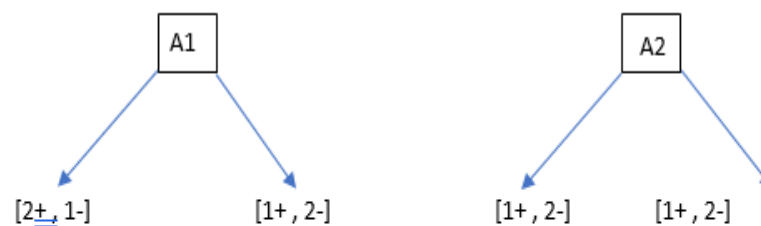
	else if (Outlook = overcast) then playtennis = yes else if (Outlook = Rain) If (wind = strong) then Playtennis = NO else Playtennis = YES Now for the day <sunny, cool, high, strong>, if it is sunny and high then donot play tennis. There is no such day like sunny—cool—high—strong Hence, player will not play.																												
4	Explain Entropy. Consider the following set of training examples. <table border="1"><thead><tr><th>Instance</th><th>Classification</th><th>A1</th><th>A2</th></tr></thead><tbody><tr><td>1</td><td>+</td><td>T</td><td>T</td></tr><tr><td>2</td><td>+</td><td>T</td><td>T</td></tr><tr><td>3</td><td>-</td><td>T</td><td>F</td></tr><tr><td>4</td><td>+</td><td>F</td><td>F</td></tr><tr><td>5</td><td>-</td><td>F</td><td>T</td></tr><tr><td>6</td><td>-</td><td>F</td><td>T</td></tr></tbody></table> i. Calculate the entropy of this collection of training examples with respect to the target function classification? ii. Calculate the information gain of A1 and A2 relative to these training examples?	Instance	Classification	A1	A2	1	+	T	T	2	+	T	T	3	-	T	F	4	+	F	F	5	-	F	T	6	-	F	T
Instance	Classification	A1	A2																										
1	+	T	T																										
2	+	T	T																										
3	-	T	F																										
4	+	F	F																										
5	-	F	T																										
6	-	F	T																										

Entropy(S)= expected number of bits needed to encode class (+ or -) of randomly drawn members of S (under the optimal, shortest length-code)
 $-p_+ \log_2 p_+ - p_- \log_2 p_-$

Gain(S,A): expected reduction in entropy due to sorting S on attribute A

$$Gain(S, A) \equiv Entropy(S) - \sum_{v \in D_A} \frac{|S_v|}{|S|} Entropy(S_v)$$

As classification is target function it is given as ,CLASSIFICATION [3+,3-]
 $E(S) = 1$ (As Classification contain equal number of positive and negative examples)



$$\begin{aligned} GAIN [S, A1] &= 1 - \left(\frac{3}{6} * E[2 + , 1 -] + \frac{3}{6} * E[1 + , 2 -] \right) \\ &= 1 - \left(\frac{3}{6} * \left(-\frac{2}{3} \log_2 \frac{2}{3} - \frac{1}{3} \log_2 \frac{1}{3} \right) + \frac{3}{6} * \left(-\frac{1}{3} \log_2 \frac{1}{3} - \frac{2}{3} \log_2 \frac{2}{3} \right) \right) \\ &= 1 - \left(\frac{3}{6} * 0.918 + \frac{3}{6} * 0.918 \right) \\ &= 0.082 \end{aligned}$$

$$\begin{aligned} GAIN [S, A2] &= 1 - \left(\frac{4}{6} * E[2 + , 2 -] + \frac{2}{6} * E[1 + , 1 -] \right) \\ &= 1 - \left(\frac{4}{6} * 1 + \frac{2}{6} * 1 \right) \\ &= 1 - 1 \\ &= 0 \end{aligned}$$

5

Suppose you are given six training points (listed in Table 1) for a classification problem with two binary attributes X1, X2, and three classes $Y \in \{1, 2, 3\}$. We will use a decision tree learner based on information gain.

Calculate the information gain for both X_1 and X_2 . You can use the approximation $\log_2 3 \approx 19/12$. Report information gains as fractions or as decimals with the precision of three decimal digits. Show your work and circle your final answers for $IG(X_1)$ and $IG(X_2)$.

Table 1: Training data for the decision tree learner

X_1	X_2	Y
1	1	1
1	1	1
1	1	2
1	0	3
0	0	2
0	0	3

★ **SOLUTION:** The equation for information gain, entropy, and conditional entropy are given by (respectively):

$$IG(X) = H(Y) - H(Y | X)$$

$$H(X) = - \sum_x \mathbf{P}(X = x) \log_2 \mathbf{P}(X = x)$$

$$H(Y | X) = \sum_x \mathbf{P}(X = x) \sum_y \mathbf{P}(Y = y | X = x) \log_2 \mathbf{P}(Y = y | X = x)$$

Using these equations we can derive the information gain for each split. First we compute the entropy $H(Y)$:

$$\begin{aligned} H(Y) &= - \sum_{y_i=1}^{n=3} \mathbf{P}(Y = y_i) \log_2 \mathbf{P}(Y = y_i) \\ &= - \sum_{y_i=1}^{n=3} \frac{1}{3} \log_2 \frac{1}{3} = \log_2 3 \approx \frac{19}{12} \end{aligned}$$

For the X_1 split we compute the conditional entropy:

$$\begin{aligned} H(Y | X_1) &= -\mathbf{P}(X_1 = 0) \sum_{y_i=1}^{n=3} \mathbf{P}(Y = y_i | X_1 = 0) \log_2 \mathbf{P}(Y = y_i | X_1 = 0) + \\ &\quad -\mathbf{P}(X_1 = 1) \sum_{y_i=1}^{n=3} \mathbf{P}(Y = y_i | X_1 = 1) \log_2 \mathbf{P}(Y = y_i | X_1 = 1) \\ &= - \left[\frac{2}{6} \left(\frac{0}{2} \log_2 \frac{0}{2} + \frac{1}{2} \log_2 \frac{1}{2} + \frac{1}{2} \log_2 \frac{1}{2} \right) + \frac{4}{6} \left(\frac{2}{4} \log_2 \frac{2}{4} + \frac{1}{4} \log_2 \frac{1}{4} + \frac{1}{4} \log_2 \frac{1}{4} \right) \right] \\ &= - \left(-\frac{2}{6} - 1 \right) \\ &= \frac{4}{3} \end{aligned}$$

DecisionTree-04-28.jpg - IrfanView (Zoom: 870 x 615)
File Edit Image Options View Help
37.2

ID3 - Capabilities and Limitations

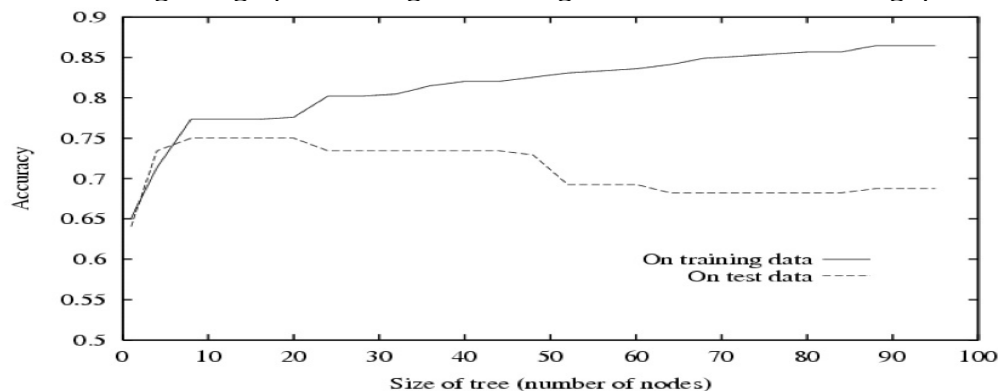
- ID3's hypothesis space of all decision trees is a **complete** space of finite discrete-valued functions.
 - Every finite discrete-valued function can be represented by some decision tree.
 - Target function is surely in the hypothesis space.
- ID3 maintains only a single current hypothesis, and outputs only a single hypothesis.
 - ID3 loses the capabilities that follow from explicitly representing all consistent hypotheses.
 - ID3 cannot determine how many alternative decision trees are consistent with the available training data.
- No backtracking on selected attributes (greedy search)
 - Local minimal (suboptimal splits)
- Statistically-based search choices
 - Robust to noisy data

Machine Learning 28

2339 x 1653 x 8 BPP 28/50 37 % 238.81 KB / 3.69 MB 8/29/2017 / 09:03:34

7

Consider the given graph showing over fitting and answer the following question.

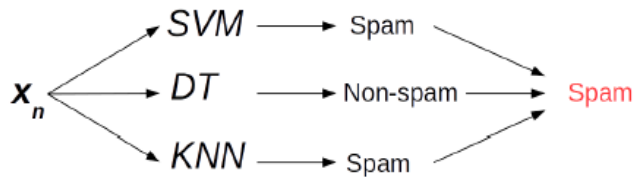


- The variance in test accuracy will increase as we increase the number of test examples.
- Given the above plot of training and test accuracy, which size decision tree would you choose to use to classify future examples? Give a one-sentence justification.
- What is the amount of over fitting in the tree you selected?
- Training error at each point on this curve provides an unbiased estimate of true error. (T/F, Justify)
- Test error at each point on this curve provides an unbiased estimate of true error. (T/F, Justify)
- Training accuracy minus test accuracy provides an unbiased estimate of the degree of over fitting. (T/F, Justify)

	g. At what number of nodes over fitting has occurred.
	a. False. The variance in test accuracy will decrease as we increase the size of the test set. b. The tree with 10 nodes. This has the highest test accuracy of any of the trees, and hence the highest expected true accuracy. c. over fitting = training accuracy minus test accuracy = $0.77 - 0.74 = 0.03$ d. False. Training error is an optimistically biased estimate of true error, because the hypothesis was chosen based on its fit to the training data. e. True. The expected value of test error (taken over different draws of random test sets) is equal to true error. f. True. We defined over fitting as test error minus training error, which is equal to training accuracy minus test accuracy. h. 20
8	With an appropriate example discuss how an attribute can be selected as best Classifier?
	A statistical property called <i>information gain</i>, measures how well a given attribute separates the training examples Information gain uses the notion of <i>entropy</i>, commonly used in information theory <i>Information gain</i> is the <i>expected</i> reduction in entropy caused by partitioning the examples on an attribute. The higher the information gain the more effective the attribute in classifying training data. Expected reduction in entropy knowing A $Gain(S, A) = Entropy(S) - \sum_{v \in Values(A)} Entropy(S_v)$ <i>Values(A)</i> possible values for A , S_v subset of S for which A has value v A1=? A2=?
9	Elaborate preference bias and restricted bias. For ex. In a popular classification algorithm, it is assumed that the entire input space can be divided into axis-parallel rectangles with each such rectangle being assigned a class label. Apply your understanding and examine whether this assumption is an example of

	preference bias or restricted bias. Support your answer with justification.
	- Preference or search biases (due to the search strategy) - ID3 searches a <i>complete</i> hypotheses space; the search strategy is <i>incomplete</i> - Restriction or language biases (due to the set of hypotheses expressible or considered) <i>Candidate-Elimination</i> searches an <i>incomplete</i> hypotheses space; the search strategy is <i>complete</i> Restriction Bias and Preference Bias - ID3 searches a complete hypothesis space - It searches incompletely through this space, from simple to complex hypotheses, until its termination condition is met - Its inductive bias is solely a consequence of the ordering of hypotheses by its search strategy. - Its hypothesis space introduces no additional bias. - Candidate Elimination searches an incomplete hypothesis space - It searches this space completely, finding every hypothesis consistent with the training data. - Its inductive bias is solely a consequence of the expressive power of its hypothesis representation. - Its search strategy introduces no additional bias. Restriction Bias and Preference Bias - The inductive bias of ID3 is a preference for certain hypothesis over others, with no hard restriction on the hypotheses that can be eventually enumerated. preference bias (search bias). - The inductive bias of Candidate Elimination is in the form of a categorical restriction on the set of hypotheses considered. restriction bias (language bias) - A <i>preference bias</i> is more desirable than a <i>restriction bias</i> because it allows the learner to work within a complete hypothesis space that is assured to contain the unknown target function. Restricted Bias: The mentioned constraints have to do with what forms the classification models can take rather than the search process to _and the specific model given some data. Hence, the above assumption is an example of restricted bias.
Unit 6	
1	When to use Deep Learning or not over others?

	1. Deep Learning outperforms other techniques if the <i>data size is large</i>. But with small data size, traditional Machine Learning algorithms are preferable. 2. Deep Learning techniques need to have <i>high end infrastructure</i> to train in reasonable time. 3. When there is <i>lack of domain understanding for feature introspection</i>, Deep Learning techniques outshines others as you have to worry less about feature engineering. 4. Deep Learning really shines when it comes to <i>complex problems such as image classification, natural language processing, and speech recognition</i>.
2	Describe the different ways to generate Diverse Learners to maximize individual accuracies and the diversity between learners?
	Different Algorithms Different learning algorithms to train different base-learners. Different algorithms make different assumptions about the data and lead to different classifiers. For example, one base-learner may be parametric, and another may be nonparametric. Different Hyper parameters We can use the same learning algorithm but use it with different hyper parameters. Examples are the number of hidden units in a multilayer perceptron, k in k-nearest neighbour, error threshold in decision trees, the kernel function in support vector machines, and so forth. With a Gaussian parametric classifier, whether the covariance matrices are shared or not are a hyper parameter. Different Input Representations Separate base-learners may be using different representations of the same input object or event, making it possible to integrate different types of sensors/measurements/modalities. Different representations make different characteristics explicit allowing better identification. Different Training Sets Another possibility is to train different base-learners by different subsets of the training set. This can be done randomly by drawing random training sets from the given sample; this is called bagging.
3	Compare and contrast Bagging and Boosting.

		Bagging	Boosting	Stacking
	Partitioning of the data into subsets	Random	Giving <u>mis</u> -classified samples higher preference	Various
	Goal to achieve	Minimize variance	Increase predictive force	Both
	Methods where this is used	Random subspace	Gradient descent	Blending
	Function to combine single models	(Weighted) average	Weighted majority vote	Logistic regression
4	What do you mean by Bagging?			
	<i>Bagging</i> is a voting method whereby base learners are made different by training them over slightly different training sets. Generating L slightly different samples from a given sample is done by bootstrap, where given a training set X of size N, we draw N instances randomly from X with <i>replacement</i>. Because sampling is done with replacement, it is possible that some instances are drawn more than once and that certain instances are not drawn at all. When this is done to generate L samples $X_j, j = 1, \dots, L$, these samples are similar because they are all drawn from the same original sample, but they are also slightly different due to chance. The base-learners d_j are trained with these L samples X_j. A learning algorithm is an <i>unstable algorithm</i> if small changes in the training set causes a large difference in the generated learner, namely, the learning algorithm has high variance. Bagging, short for bootstrap aggregating, uses bootstrap to generate L training sets, trains L base-learners using an unstable learning procedure, and then, during testing, takes an average (Breiman 1996). Bagging can be used both for classification and regression.			
5	Explain ensemble learning?			
	By combining individual models, the ensemble model tends to be more flexible (less bias) and less data-sensitive (less variance). • Better accuracy (low error) • Higher consistency (avoid over fitting) • Reduces bias and variances The different types of ensemblig techniques are: Voting or Averaging of predictions of multiple pre-trained models.  <pre> graph LR xn["x_n"] --> SVM xn --> DT xn --> KNN SVM --> Spam1[Spam] DT --> NonSpam[Non-spam] KNN --> Spam2[Spam] Spam1 --> Spam3[Spam] NonSpam --> Spam3 Spam2 --> Spam3 </pre> Stacking": Use predictions of multiple models as \features" to train a new model and use the new model to make predictions on test data.			

	<pre> graph LR x_n[x_n] --> SVM[SVM] x_n --> DT[DT] x_n --> KNN[KNN] SVM --> Spam1[Spam] DT --> NonSpam[Non-spam] KNN --> Spam2[Spam] Spam1 --> L2[Level 2 Model (e.g. DT)] NonSpam --> L2 Spam2 --> L2 L2 --> Spam3[Spam] </pre> The diagram illustrates a two-level model for spam detection. At the first level, three models (SVM, DT, and KNN) take an input x_n and produce predictions: SVM predicts 'Spam', DT predicts 'Non-spam', and KNN predicts 'Spam'. These predictions are then used as binary features to train a second-level model (e.g., DT), which finally outputs 'Spam'.
6	What is need of Reinforcement learning?
	Reinforcement Learning (RL) corrects all three drawbacks of the traditional approach. First, RL requires very little programmer effort, because most of the work is done by a process of automatic training, not programming. Second, if the environment changes, the training can be re-run, without any additional programming. In fact, retraining can be done continuously, online, as the system runs. Third, RL is mathematically guaranteed to converge to an optimal policy.
7	Outline different ways of model combination schemes.
	There are different ways the multiple base-learners are combined to generate the final output: - multi expert: methods have base-learners that work in <i>parallel</i> Combination. These methods can in turn be divided into two: - In the <i>global</i> approach, also called <i>learner fusion</i>, given an input, all base-learners generate an output, and all these outputs are used. Examples are <i>voting</i> and <i>stacking</i>. - In the <i>local</i> approach, or <i>learner selection</i>, for example, in <i>mixture of experts</i>, there is a <i>gating</i> model, which looks at the input and chooses one (or very few) of the learners as responsible for generating the output. - multistage: methods use a <i>serial</i> approach where the next combination base-learner is trained with or tested on only the instances where the previous base-learners are not accurate enough. The idea is that the base-learners (or the different representations they use) are sorted in increasing complexity so that a complex base-learner is not used (or its complex representation is not extracted) unless the preceding simpler base-learners are not confident. An example is <i>cascading</i>.